



04

Chemical bonding and structure

Essential ideas

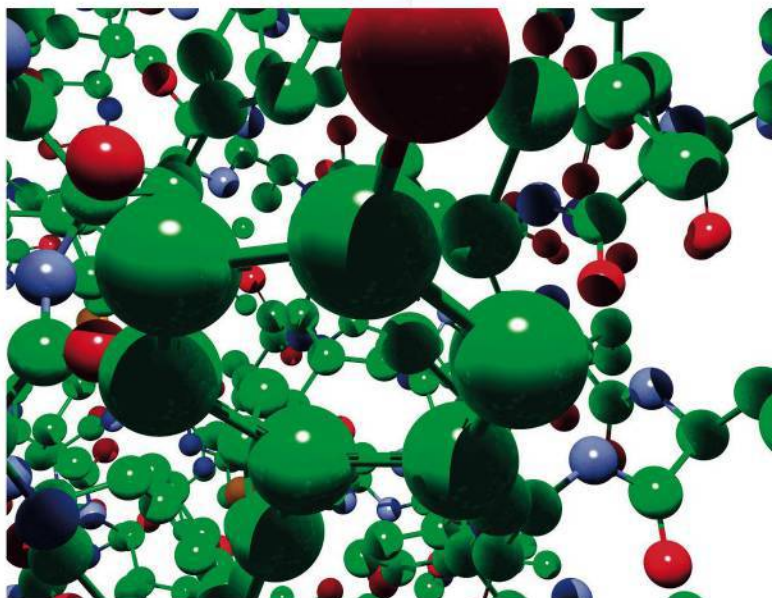
- 4.1** Ionic compounds consist of ions held together in lattice structures by ionic bonds.
- 4.2** Covalent compounds form by the sharing of electrons.
- 4.3** Lewis (electron dot) structures show the electron domains in the valence shell and are used to predict molecular shape.
- 4.4** The physical properties of molecular substances result from different types of forces between their molecules.
- 4.5** Metallic bonds involve a lattice of cations with delocalized electrons.
- 14.1** Larger structures and more in-depth explanations of bonding systems often require more sophisticated concepts and theories of bonding.
- 14.2** Hybridization results from the mixing of atomic orbitals to form the same number of new equivalent hybrid orbitals that can have the same mean energy as the contributing atomic orbitals.

Scanning electron micrograph of graphite shows its layered structure. Graphite is a form of pure carbon, in which the bonding of atoms within a layer is much stronger than the forces between the layers. An understanding of bonding enables scientists to explain and predict many of the properties of materials.

We learned in Chapter 2 that all elements are made of atoms but that there are only about 100 chemically different types of atom. Yet we know that we live in a world made up of literally millions of different substances: somehow these must all be formed from just these 100 atomic building blocks. The extraordinary variety arises from the fact that atoms readily combine with each other and they do so in a myriad of different ways. They come together in small numbers or large, with similar atoms or very different atoms, but always the result of the combination is a stable association known as a **chemical bond**. Atoms linked together by bonds therefore have very different properties from their parent atoms.

In this chapter we will study the main types of chemical bonds – the ionic bond, the covalent bond, and the metallic bond – and also consider other forces that help to hold substances together. Our study of the covalent bond at this level will use some of the concepts from quantum mechanical theory developed in Chapter 2 to explain the shapes and properties of molecules in more detail. As electrons are the key to the formation of all these bonds, a solid understanding of electron configurations will help you.

Chemical reactions take place when some bonds break and others re-form. Being able to predict and understand the nature of the bonds within a substance is therefore central to explaining its chemical reactivity.



A molecule of insulin, the hormone essential for the regulation of glucose in the body. The ball-and-stick model shows all the atoms and bonds within the protein molecule. Insulin was the first protein to have its entire structure elucidated.

4.1 Ionic bonding and structure

Understandings:

- Positive ions (cations) form by metals losing valence electrons.
- Negative ions (anions) form by non-metals gaining electrons.
- The number of electrons lost or gained is determined by the electron configuration of the atom.
- The ionic bond is due to electrostatic attraction between oppositely charged ions.
- Under normal conditions, ionic compounds are usually solids with lattice structures.

Applications and skills:

- Deduction of the formula and name of an ionic compound from its component ions, including polyatomic ions.

Guidance

Students should be familiar with the names of these polyatomic ions: NH_4^+ , OH^- , NO_3^- , HCO_3^- , CO_3^{2-} , SO_4^{2-} , and PO_4^{3-} .

- Explanation of the physical properties of ionic compounds (volatility, electrical conductivity, and solubility) in terms of their structure.

An ion is a charged particle. Ions form from atoms or from groups of atoms by loss or gain of one or more electrons.



It may help you to remember: CATion is PUSSYtive and aNion is Negative.



Ions form when electrons are transferred

All atoms are electrically neutral, even though they contain charged particles known as protons and electrons. This is because the number of protons (+) is equal to the number of electrons (−), and so their charges cancel each other out. The positively charged protons, located within the nucleus of the atom, are not transferred during chemical reactions. Electrons, however, positioned outside the nucleus, are less tightly held and outer electrons, known as **valence electrons**, can be transferred when atoms react together. When this happens the atom is no longer neutral, but instead carries an electric charge and is called an **ion**. The charge on the ion which forms is therefore determined by how many electrons are lost or gained.

We learned in Chapter 3 that the group number in the Periodic Table relates to the number of electrons in the outer shell of the atoms of all the elements in that group. We also learned that Group 18 elements, known as the **noble gases**, where the atoms all have full outer shells of electrons, are especially stable and have almost no tendency to react at all. This full outer shell behaves in a sense like the ‘ultimate goal’ for other atoms: they react to gain the stability associated with this by losing or gaining the appropriate number of electrons, whichever is the easiest (in energetic terms).



An energy-efficient light bulb. A compact fluorescent lamp has a gas-filled glass tube which contains a small amount of mercury vapour mixed with argon under low pressure. As argon is a noble gas, it is very unreactive owing to its stable electron arrangement, so it helps to provide an inert environment.

- Elements that have a small number of electrons in their outer shells (Groups 1, 2, and 13) will lose those electrons and form positive ions called **cations**. These elements are the **metals**.
- Elements that have higher numbers of electrons in their outer shells (Groups 15, 16, and 17) will gain electrons and form negative ions called **anions**. These elements are the **non-metals**.

We are now able to summarize how the position of an element in the Periodic Table enables us to predict the type of ion that it will form.

Group number	Example	Number of valence electrons	Electrons lost or gained	Number of electrons transferred	Charge on ion formed	Type of element
1	sodium	1	lost	1	1+	metal
2	calcium	2	lost	2	2+	metal
13	aluminium	3	lost	3	3+	metal
14	carbon	4	–	–	–	non-metal
15	phosphorus	5	gained	3	3–	non-metal
16	oxygen	6	gained	2	2–	non-metal
17	bromine	7	gained	1	1–	non-metal

Note that elements in Group 14, having four electrons in their outer shell, do not have a tendency to gain or to lose electrons, and so they generally do not form ions. This is because the energy involved in transferring four electrons would simply be too large to be favourable. These elements therefore react to form a different type of bond, which we will discuss later in this chapter.

Worked example

Refer to the Periodic Table to deduce the charge on the ion formed when the following elements react:

- lithium
- sulfur
- argon

Solution

- lithium is in Group 1 so forms Li^+
- sulfur is in Group 16 so forms S^{2-}
- argon is in Group 18 so does not form ions

For some elements though, it is difficult to predict the ion that will form from its position in the Periodic Table. For example, as we learned in Chapter 3, the metals occurring in the middle of the Periodic Table, known as the **transition elements**, have an electron configuration that allows them to lose different numbers of electrons from their d sub-shell and so form stable ions with different charges. The transition element iron, Fe, for example, can form Fe^{2+} by losing two electrons or Fe^{3+} by losing three electrons, depending on the reacting conditions. The two ions have distinct properties, such as forming compounds with different colours.



When an atom loses electrons it forms a positive ion, called a cation. When an atom gains electrons it forms a negative ion, called an anion. The number of charges on the ion formed is equal to the number of electrons lost or gained.

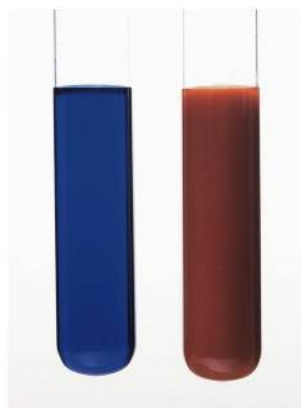


Metals form cations by losing valence electrons. Non-metals form anions by gaining electrons.



Do we have direct or indirect evidence for the existence of ions? Is there an important difference between the two types of evidence?

Compounds containing different ions of iron can be distinguished by colour: the left beaker contains $\text{Fe}^{2+}(\text{aq})$ and the right beaker $\text{Fe}^{3+}(\text{aq})$. Similar colour changes occur when iron rusts as it reacts with oxygen to form these different ions.



Fehling's reagent uses the different colours of the copper ions to test for simple sugars.

The left tube containing the blue Cu^{2+} ion changes to the red Cu^+ ion seen on the right when warmed with glucose or other 'reducing sugars'.

Likewise, the element copper can exist as Cu^{2+} and Cu^+ and again these ions can be distinguished by colour.

Other examples of elements that form ions that are not obvious from their group number are:

- lead, Pb, despite being in Group 14, forms a stable ion Pb^{2+}
- tin, Sn, also in Group 14, can form Sn^{4+} and Sn^{2+}
- silver, Ag, forms the ion Ag^+
- hydrogen, H, can form H^- (hydride) as well as the more common H^+ .

When the charge on an ion needs to be specified, the oxidation number is given in Roman numerals in brackets after the name of the element. For example, the red Cu^+ ion shown above can be written as copper(I) oxide. Oxidation numbers are explained fully in Chapter 9.

Finally, there are some ions that are made up of more than one atom which together have experienced a loss or gain of electrons and so carry a charge. These species are called **polyatomic ions**, and many of them are found in commonly occurring compounds. It will help you to become familiar with the examples in the table below, as you will often use them when writing formulas and equations. (Note that this information is not supplied in the IB data booklet.)

Polyatomic ion name	Charge on ion	Symbol	Example of compound containing this ion
nitrate	1−	NO_3^-	lead nitrate
hydroxide	1−	OH^-	barium hydroxide
hydrogencarbonate	1−	HCO_3^-	potassium hydrogencarbonate
carbonate	2−	CO_3^{2-}	magnesium carbonate
sulfate	2−	SO_4^{2-}	copper sulfate
phosphate	3−	PO_4^{3-}	calcium phosphate
ammonium	1+	NH_4^+	ammonium chloride

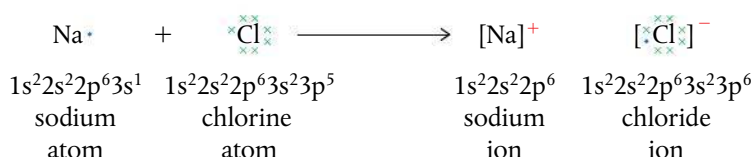
We will learn how to write the formulas for these compounds in the next section.

When writing the symbol for ions, note the charge is written as a superscript with the number first and the charge next, e.g. N^{3-} . When an ion X carries a charge of 1+ or 1− it is written just as X^+ or X^- .

Note the common names of some compounds give a clue to their composition. Here you can see that the ending '−ate' refers to ions that contain oxygen bonded to another element.

Ionic compounds form when oppositely charged ions attract

Ions do not form in isolation. Rather the process of **ionization** – where electrons are transferred between atoms – occurs when an atom that loses electrons passes them directly to an atom that gains them. Typically, this means electrons are transferred from a metal element to a non-metal element. For example, sodium (metal) passes an electron to chlorine (non-metal) when they react together.



Note that both ion products, Na^{+} and Cl^{-} , have the electron configuration of a noble gas.

Now the oppositely charged ions resulting from this electron transfer are attracted to each other and are held together by electrostatic forces. These forces are known as an **ionic bond**, and ions held together in this way are known as **ionic compounds**.

Remember that in forming the ionic compound there is no net loss or gain of electrons, and so the ionic compound, like the atoms that formed it, must be electrically neutral. Writing the formula for the ionic compound therefore involves *balancing the total number of positive and negative charges*, taking into account the different charges on each ion.

For example, magnesium oxide is made up of magnesium ions Mg^{2+} and oxide ions O^{2-} . Here each magnesium atom has transferred *two* electrons to each oxygen atom and so the compound contains equal numbers of each ion. Its formula, $\text{Mg}^{2+}\text{O}^{2-}$, is usually written as MgO . But when magnesium reacts with fluorine, each Mg loses *two* electrons, whereas each F gains only *one* electron. So it will take two fluorine atoms to combine with each magnesium atom, and the compound that results will therefore have the ratio $\text{Mg} : \text{F} = 1 : 2$. This is written as $\text{Mg}^{2+}\text{F}_2^{-}$ or MgF_2 .

Worked example

Write the formula for the compound that forms between aluminium and oxygen.

Solution

- 1 Check the Periodic Table for the ions that each element will form.
aluminium in Group 13 will form Al^{3+} ; oxygen in Group 16 will form O^{2-}
- 2 Write the number of the charge above the ion:

3	2
Al	O

Cross-multiply these numbers



Or you can directly balance the charges: here you need 6 of each charge:

$$2 \times \text{Al}^{3+} = 6+ \text{ and } 3 \times \text{O}^{2-} = 6-$$

- 3 Write the final formula using subscripts to show the number of ions: Al_2O_3



Coloured scanning electron micrograph of crystals of table salt, sodium chloride, NaCl . The very reactive elements sodium and chlorine have combined together to form this stable compound containing Na^{+} and Cl^{-} ions.



The ionic bond is due to electrostatic attraction between oppositely charged ions.



Note that in many non-metal elements the ending of the name changes to '-ide' when ions are present. For example, *chlorine* (the element) becomes *chloride* (the ion), *oxygen* becomes *oxide*, *nitrogen* becomes *nitride*, etc.



Note the convention in naming ionic compounds that the positive ion is written first and the negative ion second.



Note that the formula of the compound shows the *simplest ratio* of the ions it contains. So, for example, magnesium oxide is not Mg_2O_2 but MgO .

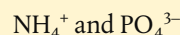
It is common practice to leave the charges out when showing the final formula. If the formula contains more than one polyatomic ion, brackets are used around the ion before the subscript.

Worked example

Write the formula for ammonium phosphate.

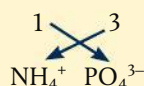
Solution

The compound contains two polyatomic ions, and you need to know these:



Balancing the charges: $3 \times \text{NH}_4^+ = 3+$ and $1 \times \text{PO}_4^{3-} = 3-$ or:

Following the steps above:



So the formula is $(\text{NH}_4)_3\text{PO}_4$.

Exercises

- Write the formula for each of the compounds in the table on page 142.
- Write the formula for each of the following compounds:

(a) potassium bromide	(d) copper(II) bromide
(b) zinc oxide	(e) chromium(III) sulfate
(c) sodium sulfate	(f) aluminium hydride
- Name the following compounds:

(a) $\text{Sn}_3(\text{PO}_4)_2$	(d) BaSO_4
(b) $\text{Ti}(\text{SO}_4)_2$	(e) Hg_2S
(c) $\text{Mn}(\text{HCO}_3)_2$	
- What are the charges on the positive ions in each of the compounds in Q3 above?
- What is the formula of the compound that forms from element A in Group 2 and element B in Group 15?
- Explain what happens to the electron configurations of Mg and Br when they react to form the compound magnesium bromide.

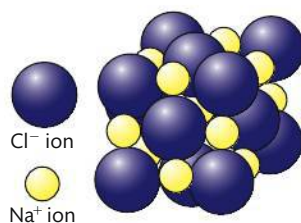


Figure 4.1 The NaCl lattice is built up from oppositely charged sodium and chloride ions.

Ionic compounds have a lattice structure

The forces of electrostatic attraction between ions in a compound cause the ions to surround themselves with ions of opposite charge. As a result, the ionic compound takes on a predictable three-dimensional crystalline structure known as an **ionic lattice**. The details of the lattice's geometry vary in different compounds, depending mainly on the sizes of the ions, but it always involves this fixed arrangement of ions based on a repeating unit. The term **coordination number** is used to express the number of ions that surround a given ion in the lattice. For example, in the sodium chloride lattice, the coordination number is six because each Na^+ ion is surrounded by six Cl^- ions and each Cl^- ion is surrounded by six Na^+ ions.

Note that the lattice consists of a very large number of ions and it can grow indefinitely. As ionic compounds do not therefore exist as units with a fixed number of ions, their formulas are simply an expression of the *ratio* of ions present. The simplest ratio is known as the **formula unit**, which is an empirical formula, as described in Chapter 1.

Make sure that you avoid the term 'molecular formula' when describing ionic compounds, but instead use the term formula unit.



We will learn in Chapter 5 that **lattice energy** is a measure of the strength of attraction between the ions within the lattice. This is greater for ions that are small and highly charged, as they have a larger charge density.

The physical properties of ionic compounds reflect their lattice structure

Physical properties are those that can be examined without chemically altering the substance. Our knowledge of ionic bonds and lattice structure helps us to interpret and explain some of these properties for ionic compounds.

Melting points and boiling points

Ionic compounds tend to have high melting and boiling points as the forces of electrostatic attraction between the ions in the lattice are strong and so require large amounts of heat energy to break. These compounds are therefore solids at room temperature, and will only melt at very high temperatures. Sodium chloride, for example, remains solid until about 800 °C.

The melting and boiling points are generally higher when the charge on the ions is greater, due to the increased attraction between the ions. For example, the table below compares the melting points of sodium oxide and magnesium oxide.

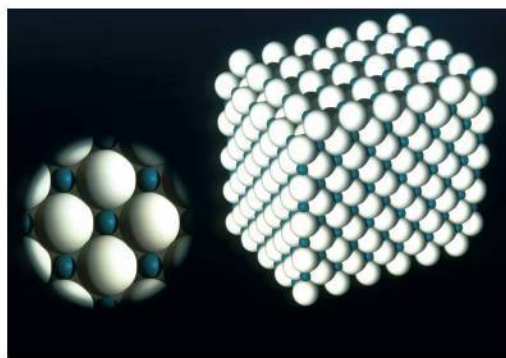
Ionic compound	Charge on metal ion	Melting point
Na ₂ O	1+	1132 °C
MgO	2+	2800 °C

The high melting points of ionic compounds become an economic consideration in many industrial processes, such as the electrolysis of molten ionic compounds discussed in Chapter 9. It can be very expensive to maintain such high temperatures, and this is an important factor in considering suitable methods to extract a reactive metal such as aluminium from its ore.

Volatility is a term used to describe the tendency of a substance to vaporize. In summary, ionic compounds can be described as having a low volatility or being non-volatile. Our everyday encounter with them as crystalline solids with low odour is consistent with this.

Solubility

Solubility refers to the ease with which a solid (the solute) becomes dispersed through a liquid (the solvent) to form a solution. You probably know from common observations that common salt, sodium chloride, readily dissolves in water but does not dissolve in oil. Why is this? There are several factors involved, but in general, solubility is determined by the degree to which the separated particles of solute are able to form bonds or attractive forces with the solvent.



Computer graphic of crystallized common salt, NaCl. Small spheres represent Na⁺ ions and larger spheres Cl⁻. The lattice is arranged so each Na⁺ ion has six oppositely charged nearest neighbours and vice versa.



Ionic liquids are efficient solvents with low volatility and are being used increasingly as solvents in Green Chemistry for energy applications and industrial processes. They are usually made of organic salts.

CHALLENGE YOURSELF

- 1 The melting point of aluminium oxide, Al₂O₃, is slightly lower than that of magnesium oxide, despite the 3+ charge on the aluminium ion. What factors might help to explain this?

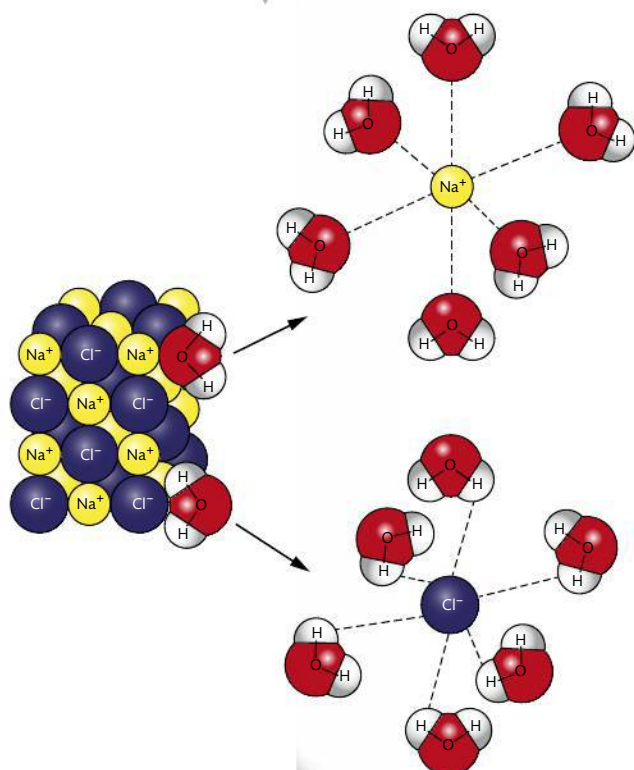


Ionic compounds have low volatility.

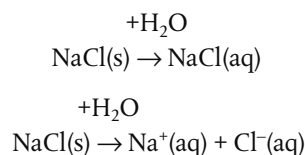


Ionic compounds are generally soluble in ionic or polar solvents but not soluble in non-polar solvents.

Figure 4.2 Dissolving of NaCl in water involves the attraction of the polar water molecules to the opposite charged ions in the NaCl lattice, and the hydration of the separated ions.



Consider an ionic compound being placed in water. As we will learn in the next section, water molecules are polar, which means they have some separation of charge in their structure. At the contact surface, the partial charges in the water molecules are attracted to ions of opposite charge in the lattice, which may cause the ions to dislodge from their position. As these ions separate from the lattice, they become surrounded by water molecules and are said to be **hydrated**. When this happens the solid is dissolved. State symbols are used to show this change as follows:



If a liquid other than water is able to dissolve the solid, the ions are said to be **solvated** and an appropriate state symbol to denote the solvent is used.

In the case of solvents like oil or hexane, C_6H_{14} , which are non-polar and so have no charge separation, there is no attraction between the liquid and the ions. So here the ions remain tightly bound to each other in the lattice, and the solid is insoluble.

This suggests that solubility trends are based on the similar chemical nature of the solute and solvent, as this is most likely to lead to successful interactions between them. The expression '*like dissolves like*' is often used to capture this notion. Note though that this is a generalized statement and somewhat over-simplified as there are some important exceptions.

Electrical conductivity

Condom conductivity test. Condoms are tested for holes by being filled with water and placed in a solution of NaCl, and then attached to electrodes. The current will not be conducted across the insulating material of the condom, but if there is a hole the current will be conducted into the salty water, triggering an alarm. All condoms are conductivity tested in this way.



will be able to move. Therefore ionic compounds as liquids or aqueous solutions do show electrical conductivity.

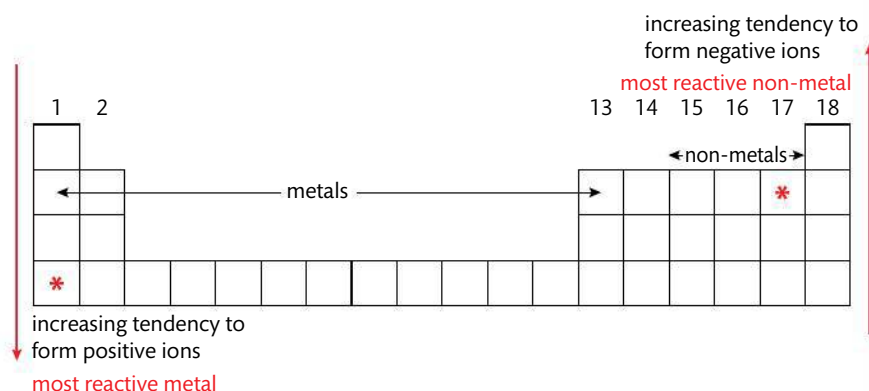
Ionic compounds do not conduct electricity in the solid state, but conduct when molten or in aqueous solution.

Brittleness

Ionic compounds are usually brittle, which means the crystal tends to shatter when force is applied. This is because movement of the ions within the lattice places ions of the same charge alongside each other, so the repulsive forces cause it to split.

Different ionic compounds have a different extent of ionic character

A **binary** compound is one that contains only two elements. We have seen that in order for any two elements to react to form an ionic compound, they must have *very different tendencies to lose or gain electrons*. A relatively simple assessment of this can be made by looking at their positions on the Periodic Table. Metals on the left lose electrons most easily, while non-metals on the right gain electrons most easily. Also, we learned in Chapter 3 that the tendency to lose electrons and form positive ions increases *down* a group, whereas the tendency to gain electrons and form negative ions increases *up* a group. So the highest tendency to react together and form ionic compounds will be between metals on the *bottom left* and non-metals on the *top right* of the Periodic Table.



Another way to judge the tendency of two elements to form an ionic compound is by looking at **electronegativity values**. As explained in Chapter 3 on page 107, electronegativity is a measure of the ability of an atom to attract electrons in a covalent bond, and is described using the Pauling scale of values. These are given in section 8 of the IB data booklet and are summarized here.

1 H 2.2																	5 B 2.0	6 C 2.6	7 N 3.0	8 O 3.4	9 F 4.0
3 Li 1.0	4 Be 1.6															13 Al 1.6	14 Si 1.9	15 P 2.2	16 S 2.6	17 Cl 3.2	
11 Na 0.9	12 Mg 1.3	19 K 0.8	20 Ca 1.0	21 Sc 1.4	22 Ti 1.5	23 V 1.6	24 Cr 1.7	25 Mn 1.6	26 Fe 1.8	27 Co 1.9	28 Ni 1.9	29 Cu 1.9	30 Zn 1.7	31 Ga 1.8	32 Ge 2.0	33 As 2.2	34 Se 2.6	35 Br 3.0			
37 Rb 0.8	38 Sr 1.0	39 Y 1.2	40 Zr 1.3	41 Nb 1.6	42 Mo 2.2	43 Tc 2.1	44 Ru 2.2	45 Rh 2.3	46 Pd 2.2	47 Ag 1.9	48 Cd 1.7	49 In 1.8	50 Sn 2.0	51 Sb 2.1	52 Te 2.1	53 I 2.7					
55 Cs 0.8	56 Ba 0.9	57 La* 1.1	72 Hf 1.3	73 Ta 1.5	74 W 1.7	75 Re 1.9	76 Os 2.2	77 Ir 2.2	78 Pt 2.2	79 Au 2.4	80 Hg 1.9	81 Tl 1.8	82 Pb 1.8	83 Bi 1.9	84 Po 2.0	85 At 2.2					
87 Fr 0.7	88 Ra 0.9	89 Ac* 1.1																			

Pauling electronegativity values

<1.0

1.0 - 1.4

1.5 - 1.9

2.0 - 2.4

2.5 - 2.9

3.0 - 4.0

Pauling
electronegativity
values

<1.0	2.0 - 2.4
1.0 - 1.4	2.5 - 2.9
1.5 - 1.9	3.0 - 4.0



NATURE OF SCIENCE

The theory that ionic compounds have a lattice structure of discrete ions leads to some predictions about their physical properties. The fact that these predictions can be tested by observation and experiment is central to the scientific process, and helps develop the theory. In general, data on the volatility, solubility, and conductivity of ionic compounds as discussed here support the lattice model. Yet there are some exceptions too, which need to be explained. The fact that aluminium oxide's melting point is not as high as would be expected from a model based purely on Al^{3+} and O^{2-} ions opens the question of what other types of bonding might be present, and why.

Figure 4.3 The pairs of elements that react most easily to form ionic compounds are metals on the bottom left of the Periodic Table and non-metals on the top right, indicated here by asterisks.

Figure 4.4 Periodic trends in electronegativity values show an increase along a period and up a group.

The extent of ionic character is determined by the *difference* in electronegativity of the bonded elements.

The determining factor in ionic bond formation is the *difference* in electronegativity values. It is generally recognized that a difference of 1.8 units or more on the Pauling scale will give a compound that is predominantly ionic.

Worked example

Explain which of the following pairs will be most likely to form an ionic bond.

- A** Be and F **B** Si and O **C** N and Cl **D** K and S

Solution

Consider the difference in electronegativity of each pair:

- A** 1.6 and 4.0 **B** 1.9 and 3.4 **C** 3.0 and 3.2 **D** 0.8 and 2.6

D has the greatest difference, so the compound K_2S will be the most ionic.

Exercises

- 7** Which fluoride is the most ionic?
A NaF **B** CsF **C** MgF_2 **D** BaF_2
- 8** Which pair of elements reacts most readily?
A $Li + Br_2$ **B** $Li + Cl_2$ **C** $K + Br_2$ **D** $K + Cl_2$
- 9** You are given two white solids and told that only one of them is an ionic compound. Describe three tests you could carry out to determine which it is.

The fact that compounds differ in their extent of ionic character suggests that other types of bonding are involved, and we will consider these in the next section. We will see that the distinction between ionic and covalent bonds is not black and white, but is best described as a **bonding continuum** with all intermediate types possible.

ionic compounds

reactive metal + reactive non-metal

electronegativity difference > 1.8

covalent compounds

two non-metals

electronegativity difference < 1.8

polar covalent compounds

$0 < \text{electronegativity difference} < 1.8$

4.2 Covalent bonding

Understandings:

- A covalent bond is formed by the electrostatic attraction between a shared pair of electrons and the positively charged nuclei.
- Single, double, and triple covalent bonds involve one, two, and three shared pairs of electrons respectively.
- Bond length decreases and bond strength increases as the number of shared electrons increases.
- Bond polarity results from the difference in electronegativities of the bonded atoms.

Guidance

Bond polarity can be shown either with partial charges, dipoles, or vectors.

Applications and skills:

- Deduction of the polar nature of a covalent bond from electronegativity values.

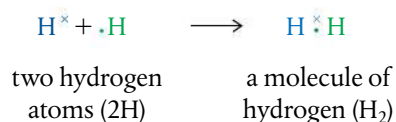
Guidance

Electronegativity values are given in section 8 of the data booklet.

A covalent bond forms by atoms sharing electrons

When atoms of two non-metals react together, each is seeking to gain electrons in order to achieve the stable electron structure of a noble gas. By sharing an electron pair they are effectively each able to achieve this. The shared pair of electrons is concentrated in the region between the two nuclei and is attracted to them both. It therefore holds the atoms together by electrostatic attraction and is known as a **covalent bond**. A group of atoms held together by covalent bonds forms a **molecule**.

For example, two hydrogen atoms form a covalent bond as follows.



Note that in the molecule H₂ each hydrogen atom has a share of two electrons so it has gained the stability of the electron arrangement of the noble gas helium, He.

The formation of the covalent bond stabilizes the atoms so energy is released as the bond forms. The forces of attraction between the nuclei and shared electrons are balanced by the forces of repulsion between the two nuclei, and this holds the atoms at a fixed distance apart. Figure 4.6 shows how the bond forms at this point of lowest energy as two hydrogen atoms approach each other and their electron density shifts.

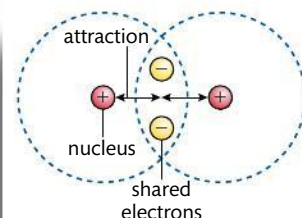
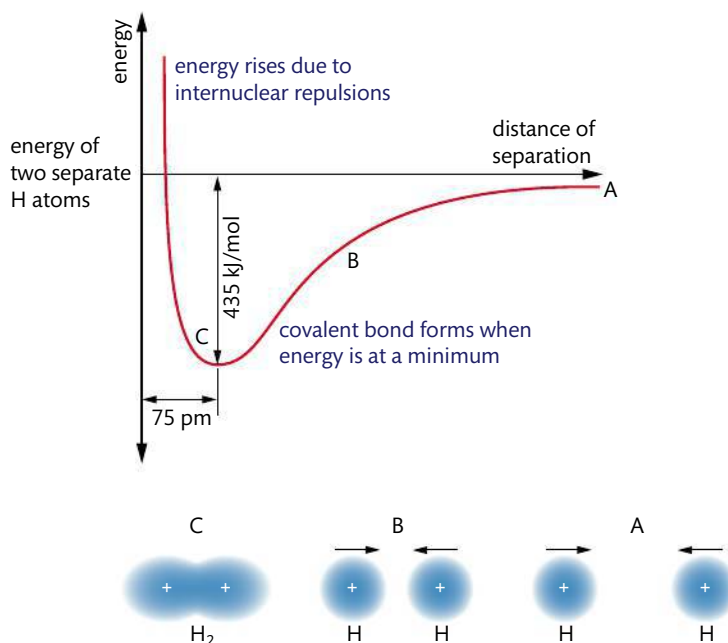
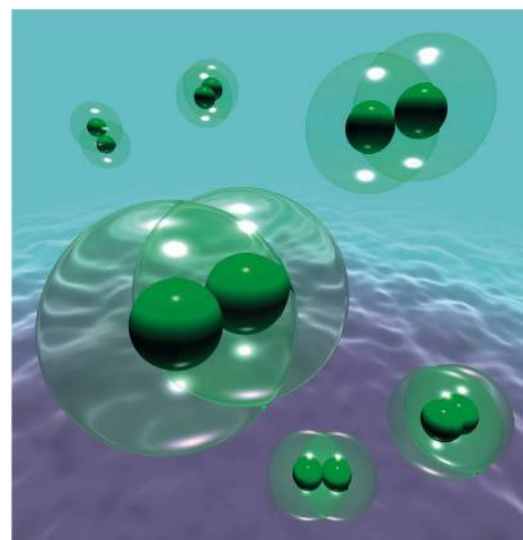


Figure 4.5 In a covalent bond the shared electrons are attracted to the nuclei of both atoms.



Computer artwork of hydrogen molecules. Each molecule consists of two hydrogen atoms bonded covalently.

Figure 4.6 When two hydrogen atoms form a covalent bond, the distance between them corresponds to the lowest energy. The letters A, B, and C show the correspondence between shifts in electron density and the distance apart of the atoms.

A covalent bond is the electrostatic attraction between a pair of electrons and positively charged nuclei. A molecule is a group of atoms held together by covalent bonds.

Molecules contain a fixed number of atoms. Those containing two atoms are described as *diatomic*, those containing three atoms are *triatomic*.

It is sometimes useful to use different symbols or colours to distinguish between the electrons of different atoms, as you see here, but in reality all electrons are the same so it is not always necessary to do this. What is crucial is to have the correct number of valence electrons in the atoms.

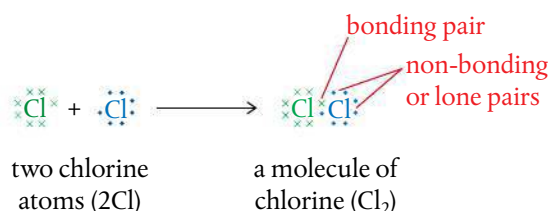
The octet rule states that when atoms react, they tend to achieve an outer shell with eight electrons.

Note the octet rule would better be considered as a guideline only, as there are exceptions – notably hydrogen, the transition elements, radicals, and atoms that form an incomplete or expanded octet.

Note that the terms *valence electrons* and *outer shell electrons* are used interchangeably. They both refer to the electrons that take part in bonding.

The triple bond in nitrogen (N_2) is difficult to break, making the molecule very stable. This is why, although nitrogen makes up about 78% of the atmosphere, it does not readily take part in chemical reactions. For example, although the element is essential for all life forms, N_2 from the atmosphere is only rarely used directly by organisms as their source of nitrogen for synthesis reactions.

Similarly, two chlorine atoms react together to form a chlorine molecule in which both atoms have gained a share of eight electrons in their outer shells (the electron arrangement of argon, Ar). This tendency of atoms to form a stable arrangement of eight electrons in their outer shell is referred to as the **octet rule**. The diagrams below only show the outer electrons, not the inner full shells, as the outer electrons are the only electrons that take part in bonding.



Note that in the chlorine molecule, each atom has three pairs of electrons that are not involved in forming the bond. These are known as **non-bonding pairs**, or **lone pairs**, and they play an important role in determining the shape of more complex molecules, as we will see later.

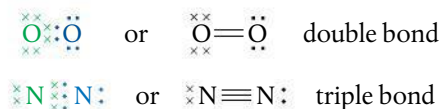
Hydrogen and chlorine also react together to form a molecule of hydrogen chloride, HCl, which has the structure shown below. Note that the formation of the covalent bond enables both H and Cl to gain a stable outer shell.



Atoms can share more than one pair of electrons to form multiple bonds

Sometimes it seems there are not enough electrons to achieve octets on all the atoms in the molecule. In these cases, the atoms will have to share more than one electron pair, in other words form a multiple bond. A **double bond** forms when two electron pairs, a total of four electrons, are shared; a **triple bond** forms when three electron pairs, a total of six electrons, are shared.

For example, the two most abundant gases in the air, oxygen and nitrogen, both exist as diatomic molecules containing multiple bonds:



As shown above, it is often convenient to use a line to represent a shared pair of electrons, two lines to represent a double bond, and three lines a triple bond. Despite this notation, we will learn in Section 14.1 (page 185) that multiple bonds contain unequal bonds. Double bonds contain one sigma bond and one pi bond, triple bonds contain one sigma bond and two pi bonds.

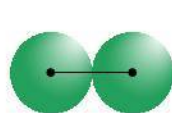
Short bonds are strong bonds

Every covalent bond is characterized by two values.

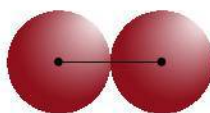
Data on both these values are given in sections 10 and 11 of the IB data booklet.

- **Bond length:** a measure of the distance between the two bonded nuclei.
- **Bond strength:** usually described in terms of **bond enthalpy**, which will be discussed in Chapter 5, and is effectively a measure of the energy required to break the bond.

As atomic radius increases as we go down a group, we would expect the atoms to form molecules with longer bonds. For example:

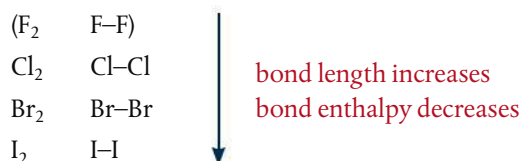


Cl₂ bond length 199 pm



Br₂ bond length 228 pm

As a result, the shared electron pair is further from the pull of the nuclei in the larger molecules, and so the bond would be expected to be weaker. Data on Group 17 elements, which exist as diatomic molecules, mostly confirm these trends.



Multiple bonds have a greater number of shared electrons and so have a stronger force of electrostatic attraction between the bonded nuclei. This means there is a greater pulling power on the nuclei, bringing them closer together, resulting in bonds that are shorter and stronger than single bonds. We can see this by comparing different bonds involving the same atoms, such as in the hydrocarbons.

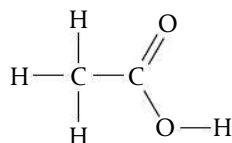
Hydrocarbon	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂
Structural formula	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} $	$ \begin{array}{c} \text{H} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{H} \end{array} $	$\text{H}-\text{C}\equiv\text{C}-\text{H}$
Bond between carbon atoms	single	double	triple
Bond length / pm	154	134	120
Bond enthalpy / kJ mol ⁻¹	346	614	839

These data confirm that triple bonds are stronger and shorter than double bonds, which are shorter and stronger than single bonds. Note though that the double bond is not twice as strong as the single bond, due to the fact that it is made up of two different bonds – one sigma bond and one pi bond that is weaker. These bonds will be described more fully in Section 14.1.

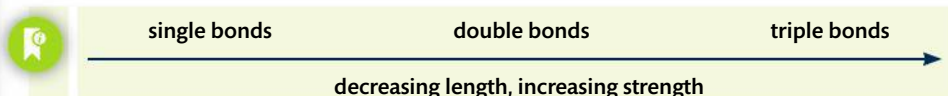
CHALLENGE YOURSELF

- Fluorine, F₂, is shown in brackets opposite, as a close look at its bond enthalpy data shows that it is an outlier in the trends described here. What explanation might account for this?

Also, we can compare two different carbon–oxygen bonds within the molecule ethanoic acid, CH_3COOH .



	C—O	C=O
Bond length / pm	143	122
Bond enthalpy / kJ mol^{-1}	358	804



Polar bonds result from unequal sharing of electrons

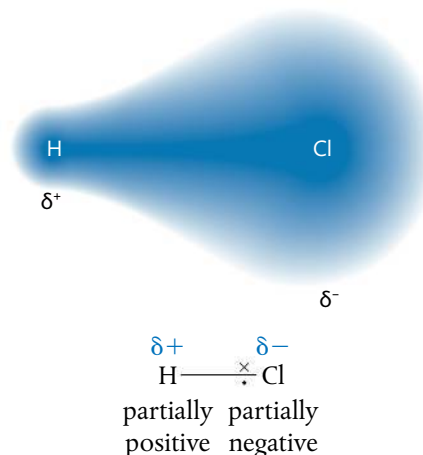
Not all sharing is equal. It may be the case that you share this textbook with a class mate, but if you have it for more than half the time, it clearly belongs more to you than to your friend. So it is with electron pairs. In simple terms, if they spend more time with one atom than the other, they are not equally shared. This occurs when there is a difference in the electronegativities of the bonded atoms as the more electronegative atom exerts a greater pulling power on the shared electrons and so gains more 'possession' of the electron pair. The bond is now unsymmetrical with respect to electron distribution and is said to be **polar**.

The term **dipole** is often used to indicate the fact that this type of bond has two separated opposite electric charges. The more electronegative atom with the greater share of the electrons, has become partially negative or δ^- , and the less electronegative atom has become partially positive or δ^+ .

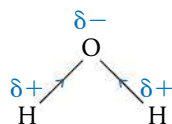
For example, in HCl the shared electron pair is pulled more strongly by the Cl atom than the H atom, resulting in a polar molecule.

Figure 4.7 The distribution of electrons in the H–Cl bond is unsymmetrical, with a greater electron density over the Cl atom.

Note that the symbol δ (Greek letter, pronounced 'delta') is used to represent a partial charge. This has no fixed value, but is always less than the unit charge associated with an ion such as X^+ or X^- .



In water, the electronegativity difference results in polar O–H bonds with the electron density greater on the oxygen.



The extent of polarity in a covalent bond varies, depending on how big a difference exists in the electronegativity values of the two bonded atoms. We can estimate this from knowledge of the periodic trends in electronegativity, discussed in Chapter 3 and summarized here.

electronegativity increases across a period

electronegativity increases up a group

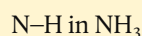
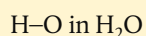
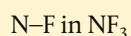
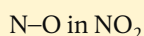
As fluorine is in the top right of the Periodic Table, it is clearly the most electronegative atom, so we can predict it will have the greater electron density whenever it is bonded to another element. Fluorine has a value of 4.0 on the Pauling scale of electronegativity given on page 147, and all other elements have lower values. This scale can be used to assess the relative polarity of a bond. For example, the table below compares the polarities of the hydrogen halides.

	Electronegativity values of atoms		Difference in electronegativity
H-F	2.2	4.0	1.8
H-Cl	2.2	3.2	1.0
H-Br	2.2	3.0	0.8
H-I	2.2	2.7	0.5

increasing bond polarity

Worked example

Use the electronegativity values on page 147 to put the following bonds in order of decreasing polarity.



Solution

bond polarity: H-O > N-F > N-H > N-O

difference in electronegativity values: 1.2 1.0 0.8 0.4

A microwave oven generates microwaves that have lower frequencies and longer wavelengths than visible light. Microwaves are used for cooking because certain wavelengths are absorbed by polar molecules, notably water, and also by sugar and fat. The absorption causes the water to heat up and cook the food, leaving the oven surface and containers of glass, plastic, or ceramic unaffected.

The only bonds that are truly non-polar are bonds between the same atoms such as the bonds in F₂, H₂, and O₂, because clearly here the *difference* in electronegativity is zero. These are sometimes referred to as **pure covalent** to express this. All other bonds have



There are several ways to denote the polar nature of a bond in a structure. One is to write the partial charges δ+ and δ- over the less electronegative and more electronegative atom respectively. Another is to represent the direction of electron pull as a vector with an arrow on the bond indicating the pull on the electrons by the more electronegative atoms. The dipole notation $\rightarrow$ is also useful. As you can see in the diagram of water here, it is quite acceptable to show more than one way together.

CHALLENGE YOURSELF

- 3 Oxygen is a very electronegative element with a value of 3.4 on the Pauling scale. Can you determine the formula of a compound in which oxygen would have a partial positive charge?

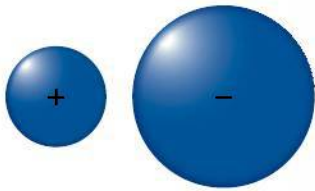


In a covalent bond, the greater the difference in the electronegativity values of the atoms, the more polar the bond.

some degree of polarity, although it may be very slight. The C–H bond, ubiquitous in organic chemistry, is often considered to be largely non-polar, although in fact carbon is slightly more electronegative than hydrogen. As we will see in Chapter 10, the low polarity of the C–H bond is an important factor in determining the properties of organic compounds.

The presence of polar bonds in a molecule has a significant effect on its properties. Effectively, the partial separation of charges introduces some ionic nature into covalent bonds, so the more polar the bond, the more like an ionic compound the molecule behaves. Polar bonds are therefore considered to be intermediates in relation to pure covalent bonds and ionic bonds, part of the bonding continuum that was mentioned on page 148. We now see why the boundaries between ionic and covalent bonds are somewhat ‘fuzzy’, and why it is often appropriate to describe substances using terms such as ‘predominantly’ covalent or ionic, or of being ‘strongly’ polar.

The idea of the bonding continuum is summarized below.

Ionic	Polar covalent	Pure covalent
complete transfer of electrons	partial transfer of electrons unequal sharing of electrons	equal sharing of electrons
Na^+Cl^-	HF, HCl, HBr, HI	Cl_2
ionic lattice		discrete molecules
		



NATURE OF SCIENCE

Scientists observe the natural world where variations in structure or properties are often continuous rather than discontinuous. In other words, a spectrum of intermediates exists and this can make classification difficult. At the same time, the properties of such intermediates can strengthen our understanding. For example, we have seen here that the existence of polar covalent molecules in the bonding continuum adds to our interpretation of electron behaviour in bond formation.

Exercises

- Which substance contains only ionic bonds?
A NaNO_3 **B** H_3PO_4 **C** NH_4Cl **D** CaCl_2
- Which of the following molecules contains the shortest bond between carbon and oxygen?
A CO_2 **B** H_3COCH_3 **C** CO **D** CH_3COOH
- For each of these molecules, identify any polar bonds and label them using $\delta+$ and $\delta-$ appropriately:
(a) HBr **(b)** CO_2 **(c)** ClF
(d) O_2 **(e)** NH_3
- Use the electronegativity values in Section 8 of the IB data booklet to predict which bond in each of the following pairs is more polar.
(a) C–H or C–Cl **(b)** Si–Li or Si–Cl **(c)** N–Cl or N–Mg

4.3 Covalent structures

Understandings:

- Lewis (electron dot) structures show all the valence electrons in a covalently bonded species.

Guidance

- The term 'electron domain' should be used in place of 'negative charge centre'.
- Electron pairs in a Lewis (electron dot) structure can be shown as dots, crosses, a dash, or any combination.
- Coordinate covalent bonds should be covered.
- The 'octet rule' refers to the tendency of atoms to gain a valence shell with a total of 8 electrons.
- Some atoms, like Be and B, might form stable compounds with incomplete octets of electrons.
- Resonance structures occur when there is more than one possible position for a double bond in a molecule.
- Shapes of species are determined by the repulsion of electron pairs according to VSEPR theory.
- Carbon and silicon form giant covalent/network covalent/macromolecular structures.

Guidance

Allotropes of carbon (diamond, graphite, graphene, C_{60} buckminsterfullerene) and SiO_2 should be covered.

Applications and skills:

- Deduction of Lewis (electron dot) structure of molecules and ions showing all valence electrons for up to four electron pairs on each atom.
- The use of VSEPR theory to predict the electron domain geometry and the molecular geometry for species with two, three, and four electron domains.
- Prediction of bond angles from molecular geometry and presence of non-bonding pairs of electrons.
- Prediction of molecular polarity from bond polarity and molecular geometry.
- Deduction of resonance structures, including C_6H_6 , CO_3^{2-} , and O_3 .
- Explanation of the properties of giant covalent compounds in terms of their structures.

Lewis diagrams are used to show the arrangement of electrons in covalent molecules

When describing the structure of covalent molecules, the most convenient method is known as a **Lewis structure**.

This uses a simple notation of dots and crosses to represent the outer energy level, or valence shell, electrons of all the atoms in the molecule. It can be derived as follows.



Gilbert Newton Lewis (1875–1946) was one of the greatest and most influential American chemists of the last century. He formulated the idea that the covalent bond consists of a shared pair of electrons, proposed the electron pair theory of acid/base reactions, and was the first person to produce a pure sample of deuterium oxide (heavy water). He also published important theories on chemical thermodynamics and first coined the term 'photon' for the smallest unit of radiant energy. Since 1916 his dot diagrams for covalent structures have been used almost universally.

Steps to follow in drawing a Lewis structure.

1 Calculate the total number of valence electrons in the molecule by multiplying the number of valence electrons of each element by the number of atoms of the element in the formula and totalling these.

2 Draw the skeletal structure of the molecule to show how the atoms are linked to each other.

4 Add more electron pairs to complete the octets (8 electrons) around the atoms (other than hydrogen which must have 2 electrons and the exceptions noted below).

3 Use a pair of crosses, a pair of dots, or a single line to show one electron pair and put a pair in each bond between atoms.

5 If there are not enough electrons to complete the octets, form double bonds and if necessary triple bonds.

6 Check that the total number of electrons in your finished structure is equal to your calculation in the first step.

Electron pairs can be represented by dots, crosses, a combination of dots and crosses, or by a line. Use whichever notation you prefer, but be prepared to recognize all versions. The important thing is to be clear and consistent, and to avoid vague and scattered dots.



Worked example

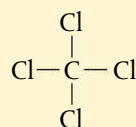
Draw the Lewis structure for the molecule CCl_4 .

Solution

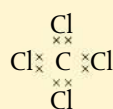
Note that in Lewis structures, as in the examples on page 157, only the valence electrons are shown, as these are the only ones that take part in bonding.

1 Total number of valence electrons = $4 + (7 \times 4) = 32$

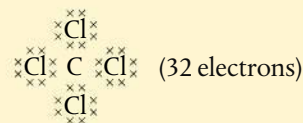
2 Skeletal structure



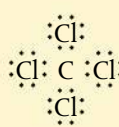
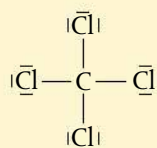
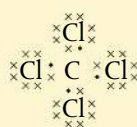
3 Bonded pairs



4 Completed Lewis structure



There are several different ways of drawing Lewis structures, and some alternative acceptable forms are shown below.



The table on page 157 shows some examples of molecules with their Lewis structures.

Molecule	Total number of valence electrons	Lewis structure
CH ₄	4 + (1 × 4) = 8	$\begin{array}{c} \text{H} \\ \times \\ \times \\ \text{H} \cdot \text{C} \cdot \text{H} \\ \times \\ \times \end{array} \quad \text{H} - \text{C} - \text{H}$
NH ₃	5 + (1 × 3) = 8	$\begin{array}{c} \times \times \\ \times \times \\ \text{H} \cdot \text{N} \cdot \text{H} \\ \times \times \\ \times \times \end{array} \quad \text{H} - \text{N} - \text{H}$
H ₂ O	(1 × 2) + 6 = 8	$\text{H} \cdot \text{O} \cdot \text{H} \quad \text{H} - \text{O} - \text{H}$
CO ₂	4 + (6 × 2) = 16	$\begin{array}{c} \times \times \\ \times \times \\ \text{O} \cdot \text{C} \cdot \text{O} \\ \times \times \\ \times \times \end{array} \quad \text{O} = \text{C} = \text{O}$
HCN	1 + 4 + 5 = 10	$\text{H} \cdot \text{C} \cdot \text{N} \cdot \quad \text{H} - \text{C} \equiv \text{N}$

When drawing the Lewis structure of ions, note that you must:

- calculate valence electrons as above and then add one electron for each negative charge and subtract one electron for each positive charge;
- put the Lewis structure in a square bracket with the charge shown outside.

Worked example

Draw the Lewis structures for:

(i) OH⁻

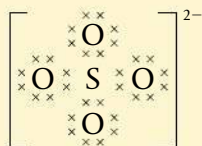
(ii) SO₄²⁻

Solution

(i) valence electrons = 6 + 1 + 1 = 8



(ii) valence electrons = 6 + (6 × 4) + 2 = 32

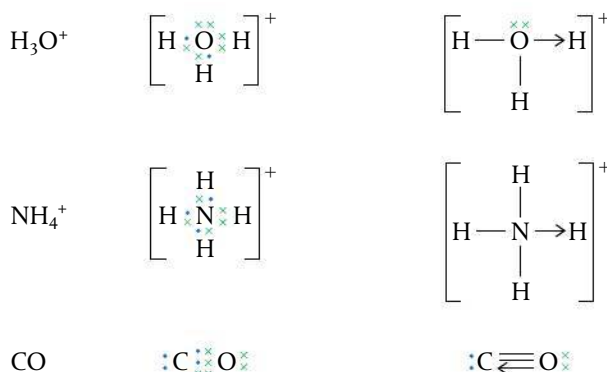


When drawing the Lewis structure of an ion, make sure you remember to put a square bracket around the structure with the charge shown clearly outside the bracket.

In coordinate bonds both shared electrons come from one atom

The examples so far involve covalent bonds where each bonded atom contributes one electron to the shared pair. However, sometimes the bond forms by *both* the electrons in the pair originating from the same atom. This means that the other atom accepts and gains a share in a donated electron pair. Such bonds are called **coordinate bonds**, also known as **dative bonds**. An arrow on the head of the bond is sometimes used to show a coordinate bond, with the direction indicating the origin of the electrons.

For example:



A coordinate bond is a covalent bond in which both the shared electrons are provided by one of the atoms.



Note that in CO the triple bond consists of two bonds that involve sharing an electron from each atom, and the third bond is a coordinate bond where both electrons come from the oxygen. The three bonds are nonetheless identical to each other. This illustrates an important point about coordinate bonds: once they are formed, they are no different from other covalent bonds. Understanding the origin of bonding electrons can be important, however, in interpreting some reaction mechanisms, such as Lewis acid/base behaviour, as we will learn in Chapter 8.

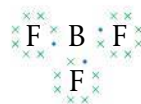
The octet rule is not always followed

There are a few molecules that are exceptions to the octet rule. Small atoms such as beryllium (Be) and boron (B) form stable molecules in which the central atom has fewer than eight electrons in its valence shell. This is known as an **incomplete octet**. For example, the structures of BeCl_2 and BF_3 are shown below.

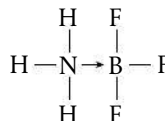
BeCl_2 valence electrons = $2 + (7 \times 2) = 16$



BF_3 valence electrons = $3 + (7 \times 3) = 24$



Molecules with incomplete octets are said to be **electron deficient**, and have a tendency to accept an electron pair from a molecule with a lone pair, such as NH_3 or H_2O . This leads to the formation of a coordinate compound in which the central atom has now gained an octet.



BF_3 is an important catalyst in several synthetic reactions as a result of this tendency to accept electrons.

Remember to include *all lone pairs* in your Lewis structures. Structures which show only bonded pairs, e.g. $\text{H}-\text{Cl}$, are structural formulas, not Lewis structures.



Exercises

14 Draw the Lewis structures of:

- | | | | |
|----------------------------|----------------------------|----------------------------|--------------------|
| (a) HF | (b) CF_3Cl | (c) C_2H_6 | (d) PCl_3 |
| (e) C_2H_4 | (f) C_2H_2 | | |

15 How many valence electrons are in the following molecules?

- | | | | |
|---------------------|--------------------|--------------------|-------------------|
| (a) BeCl_2 | (b) BCl_3 | (c) CCl_4 | (d) PH_3 |
| (e) SiCl_2 | (f) NCl_3 | | |

16 Use Lewis structures to show the formation of a coordinate bond between H_2O and H^+ .

17 Draw the Lewis structures of:



Molecules that have an expanded octet have central atoms that are larger and are found in Period 3 (and beyond), where the valence shell can hold 18 electrons. Their structures, involving five and six electron pairs, are discussed in Section 14.1.

VSEPR theory: The shape of a molecule is determined by repulsion between electron domains

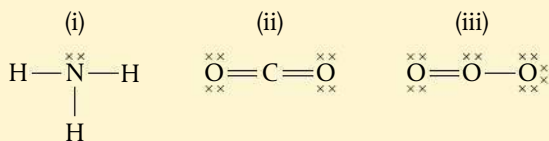
Once we know the Lewis structure of a molecule, we can predict exactly how the bonds will be orientated with respect to each other in space – in other words, the three-dimensional shape of the molecule. This is often a crucial feature of a substance in determining its reactivity. For example, biochemical reactions depend on a precise ‘fit’ between the enzyme, which controls the rate of the reaction, and the reacting molecule, known as the substrate. Anything that changes the shape of either of these may therefore alter the reaction dramatically; many drugs work in this way.

Predictions of molecular shape are based on the **Valence Shell Electron Pair Repulsion (VSEPR) theory**. As its name suggests, this theory is based on the simple notion that *because electron pairs in the same valence shell carry the same charge, they repel each other and so spread themselves as far apart as possible*.

Before applying this theory, it is useful to clarify some of the language involved because the term VSEPR, evidently referring to Electron Pairs, is actually an oversimplification. As we have seen, molecules frequently contain multiple pairs of shared electrons, and these behave as a single unit in terms of repulsion because they are orientated together. So a better, more inclusive, term than electron pair is **electron domain**. This includes all electron locations in the valence shell, be they occupied by lone pairs, single, double, or triple bonded pairs. What matters in determining shape is the *total number of electron domains*, and this can be determined from the Lewis structure.

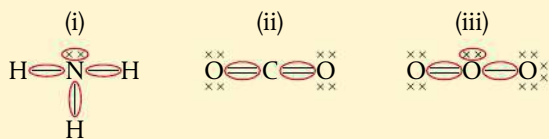
Worked example

How many electron domains exist in the central atom of the following molecules whose Lewis structures are shown?



Solution

Each electron domain is ringed in red.



(i) 4 electron domains: 3 bonding and 1 non-bonding

(ii) 2 electron domains: 2 bonding and 0 non-bonding

(iii) 3 electron domains: 2 bonding and 1 non-bonding



Note that hydrogen, H, which forms only one bond, must always be in a terminal position and can never be a central atom in a molecule.

TOK

People often find this topic easier to understand when they can build and study models of the molecules in three dimensions. Does this suggest different qualities to the knowledge we acquire in different ways?

Technically then, VSEPR theory should be called Valence Shell Electron Domain Repulsion. It can be summarized as follows:

- The repulsion applies to electron domains, which can be single, double, or triple bonding electron pairs, or non-bonding pairs of electrons.
- The total number of electron domains around the central atom determines the geometrical arrangement of the electron domains.
- The shape of the molecule is determined by the angles between the bonded atoms.
- Non-bonding pairs (lone pairs) have a higher concentration of charge than a bonding pair because they are not shared between two atoms, and so cause slightly more repulsion than bonding pairs. The repulsion decreases in the following order:

lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair

As a result, molecules with lone pairs on the central atom have some distortions in their structure that reduce the angle between the bonded atoms.

Note that in the diagrams in this section, non-bonding electrons on the surrounding atoms have been omitted for clarity – these are therefore *not* Lewis structures.

Note that when you are asked to draw a Lewis structure, you must show all the non-bonding pairs on all the atoms.



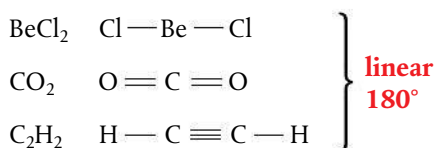
Species with two electron domains

Molecules with two electron domains will position them at **180°** to each other, giving a **linear** shape to the molecule.



The repulsion between electron pairs varies as follows:

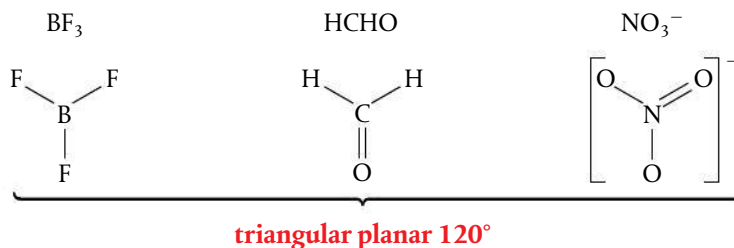
strongest repulsion	↑	lone pair–lone pair
		lone pair–bonding pair
weakest repulsion		bonding pair–bonding pair



Species with three electron domains

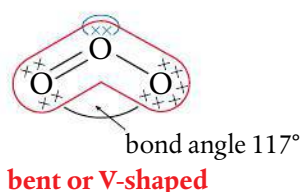
Molecules with three electron domains will position them at **120°** to each other, giving a **triangular planar** shape to the electron domain geometry. If all three electron domains are bonding, the shape of the molecule will also be triangular planar.

The electron domain geometry is determined by the positions of *all* the electron domains, but the molecular geometry depends only on the positions of the *bonded* atoms.



However, if one of the electron domains is a lone pair, this will not be 'seen' in the overall shape of the molecule, as it is part of the central atom. The molecular shape is determined by the positions of the atoms fixed by the bonding pairs only. A further consideration is that, as described above, lone pairs cause slightly more repulsion than bonding pairs, so in their presence the angles are slightly altered.

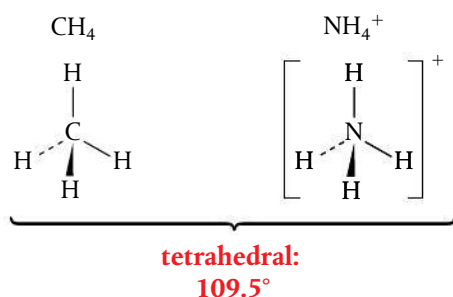
For example, as shown on page 159, the Lewis structure of ozone, O_3 , has three electron domains. Therefore the electron domain geometry of the molecule is triangular planar. But, as only two of the three electron domains are bonding electrons, the molecular geometry will not be triangular planar. The red outline below shows the positions of the bonding electrons that determine the shape. The molecule is described as **bent** or **V-shaped**. The lone pair of electrons distorts the shape slightly, so the angle is slightly less than 120° ; it is approximately 117° .



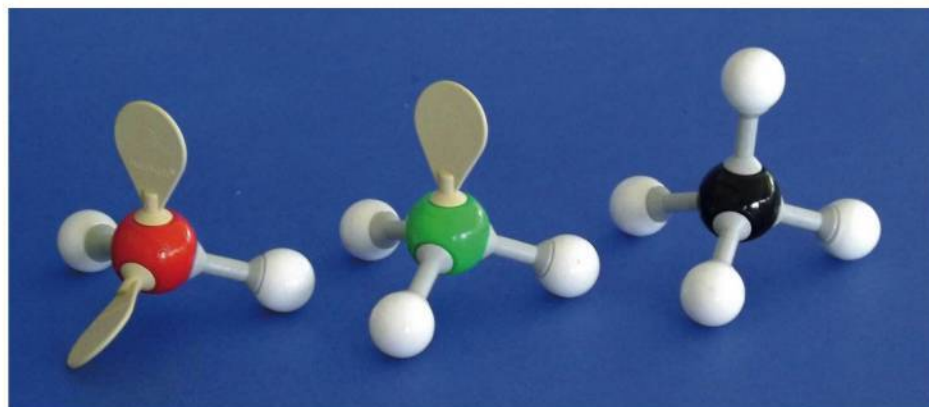
If there are lone pairs on the central atom, the electron domain geometry and the molecular geometry will not be the same as each other.

Species with four electron domains

Molecules with four electron domains will position them at 109.5° to each other, giving a **tetrahedral** shape to the electron domains. If all four electron domains are bonding, the shape of the molecule will also be tetrahedral.



However, if one or more of the electron domains is a lone pair, we must again focus on the number of bonded pairs to determine the shape of the molecule. The table on page 162 compares molecules with 0, 1, and 2 lone pairs respectively.



Models of H_2O , NH_3 , and CH_4 showing the different molecular geometry due to the different numbers of lone pairs on the central atom.

	CH ₄	NH ₃	H ₂ O
Lewis structure			
Number of electron domains	4	4	4
Electron domain geometry	tetrahedral	tetrahedral	tetrahedral
Number of lone pairs	0	1	2
Number of bonded electron domains	4	3	2
Molecular geometry	tetrahedral 	trigonal pyramidal 	bent or V-shaped
Bond angles	109.5°	approximately 107°	approximately 105°

Note that the presence of two lone pairs in H₂O causes more repulsion than the single lone pair in NH₃. This is why the bond angle is reduced further in H₂O from the 109.5° of the symmetrical tetrahedron.

Here is a summary of the steps used in determining the shape of a molecule.

- 1 Draw the Lewis structure, following the steps on page 156.
- 2 Count the total number of electron domains on the central atom.
- 3 Determine the electron domain geometry as follows:
2 electron domains → linear
3 electron domains → triangular planar
4 electron domains → tetrahedral
- 4 Determine the molecular geometry from the number of bonding electron domains.
- 5 Consider the extra repulsion caused by the lone pairs and adjust the bond angles accordingly.

Always draw the Lewis structure before attempting to predict the shape of a molecule, as you have to know the total number of bonding pairs and lone pairs around the central atom.



With 4 electron domains on the central atom: 0 lone pairs → tetrahedral, 1 lone pair → trigonal pyramid, 2 lone pairs → V-shaped.



Exercises

18 Predict the shape and bond angles of the following molecules:

- | | | | |
|----------------------|------------------------|---------------------|---------------------|
| (a) H ₂ S | (b) CF ₄ | (c) HCN | (d) NF ₃ |
| (e) BCl ₃ | (f) NH ₂ Cl | (g) OF ₂ | |

19 Predict the shape and bond angles of the following ions:

- | | | | |
|-----------------------------------|------------------------------------|----------------------------------|----------------------------------|
| (a) CO ₃ ²⁻ | (b) NO ₃ ⁻ | (c) NO ₂ ⁺ | (d) NO ₂ ⁻ |
| (e) ClF ₂ ⁺ | (f) SnCl ₃ ⁻ | | |

20 How many electron domains are there around the central atom in molecules that have the following shapes?

- | | | |
|------------------------|-----------------------|------------|
| (a) tetrahedral | (b) bent | (c) linear |
| (d) trigonal pyramidal | (e) triangular planar | |

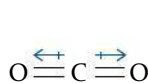
Molecules with polar bonds are not always polar

We have learned that the polarity of a *bond* depends on the charge separation between its two bonded atoms, which is a result of differences in their electronegativities. The polarity of a *molecule*, however, depends on:

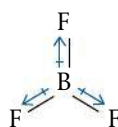
- the polar bonds that it contains;
- the way in which such polar bonds are orientated with respect to each other, in other words, on the shape of the molecule.

If the bonds are of equal polarity (i.e. involving the same elements) *and* are arranged symmetrically with respect to each other, their charge separations will oppose each other and so will effectively cancel each other out. In these cases the molecule will be non-polar, despite the fact that it contains polar bonds. It is a bit like a game of tug-of-war between players who are equally strong and symmetrically arranged.

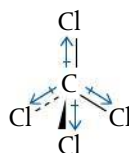
The molecules below are all non-polar because the dipoles cancel out. The arrows, shown here in blue, are the notation for a dipole that results from the pull of electrons in the bond towards the more electronegative atom.



CO₂



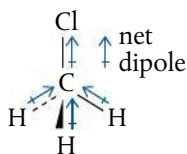
BF₃



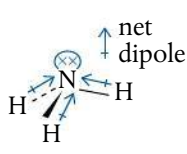
CCl₄

However, if *either* the molecule contains bonds of different polarity, *or* its bonds are not symmetrically arranged, then the dipoles will not cancel out, and the molecule will be polar. Another way of describing this is to say that it has a **net dipole moment**, which refers to its turning force in an electric field. This is what would happen in a game of tug-of-war if the players were not equally strong or were not pulling in exactly opposite directions.

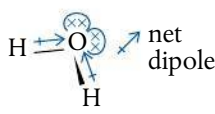
The molecules below are all polar because the dipoles do not cancel out.



CH₃Cl



NH₃



H₂O

Exercises

21 Predict whether the following will be polar or non-polar molecules:

(a) PH₃

(b) CF₄

(c) HCN

(d) BeCl₂

(e) C₂H₄

(f) ClF

(g) F₂

(h) BF₃



▲ VSEPR theory in nature. Different numbers of snowberries growing together spontaneously adopt the geometry of maximum repulsion. Clockwise from the top: four berries form a tetrahedral shape bond angle 109.5°, three berries form a triangular planar shape bond angle 120°, and two berries are linear bond angle 180°. What other examples of VSEPR shapes can you find in nature?



▲ **Figure 4.8** Equal and opposite pulls cancel each other out.



▲ **Figure 4.9** When the pulls are not equal and opposite there is a net pull.

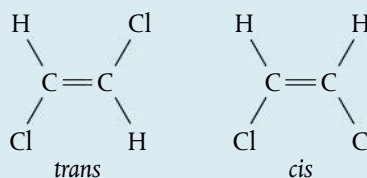


Bond dipoles can be thought of as vectors, so the polarity of a molecule is the resultant force. This approach may help you to predict the overall polarity from considering the sum of the bond dipoles.

CHALLENGE YOURSELF

- 4 If you were given two solutions and told one contained a polar substance and the other a non-polar substance, suggest experiments that might enable you to identify which is which.

22 The molecule $C_2H_2Cl_2$ can exist as two forms known as *cis-trans* isomers, which are shown below.



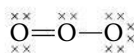
Determine whether either of these has a net dipole moment.

Electrons in multiple bonds can sometimes spread themselves between more than one bonding position

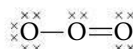
Our account of covalent bonding so far describes a bond as a pair of electrons held in a specific position within a molecule. However, in some molecules bonding electrons are less restricted than this. Instead of being confined to one location they show a tendency to be shared between more than one bonding position, and are said to be **delocalized**. Free from the constraints of a single bonding position, delocalized electrons spread themselves out, giving greater stability to the molecule or ion.

Delocalization is a characteristic of electrons in multiple bonds when there is more than one possible position for a

double bond within a molecule. For example, let us look again at the Lewis structure of ozone, O_3 , from page 159,



We can see that a different Lewis structure would be equally valid:



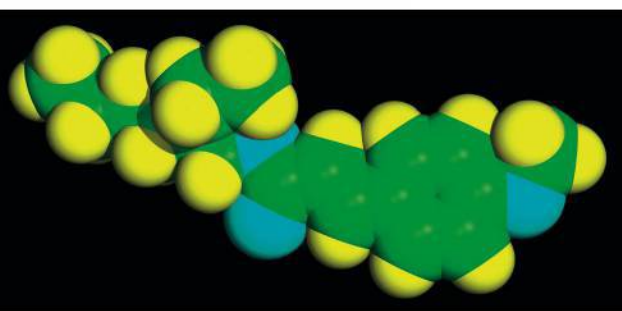
These structures suggest that the molecule should contain one oxygen–oxygen double bond and one oxygen–oxygen single bond, which we would expect to be different in bond length and bond strength. However, experimental data reveal that ozone actually contains two *equal* oxygen–oxygen bonds, intermediate in length and strength between single and double bonds.

	O–O single bond	O=O double bond	Oxygen–oxygen bonds in O_3
Bond length / pm	148	121	127
Bond enthalpy / kJ mol^{-1}	144	498	364

We can explain this by considering that the electrons from the double bond have delocalized and spread themselves equally between both possible bonding positions. The true structure for ozone is a blend or hybrid of the Lewis structures shown above.



This cannot be accurately represented by a single Lewis structure, so a concept known as **resonance** is introduced. The Lewis structures that can be drawn are known as



Computer graphic of a molecule of the sunscreen $C_{18}H_{26}O_3$. Carbon atoms are shown in green, hydrogen atoms in yellow, and oxygen atoms in blue. Note the benzene ring and carbon–carbon double bonds in the molecule. Sunscreens help to protect the skin from sun damage by absorbing UV radiation and releasing it as heat. They are able to do this due to the many double bonds and delocalized electrons in their structure.

resonance structures, depicted as shown below, while it is recognized that neither of these is the correct structure. The true structure is an intermediate form, known as the **resonance hybrid**.



It is as if you were trying to explain to someone what a mule (a cross between a horse and a donkey) looks like: you might draw a horse and a donkey and a double-headed arrow between them implying the mule is an intermediate. You wouldn't mean that the animal is a horse one minute and a donkey the next! Likewise, resonance structures do not represent forms that flip from one to the other. Resonance structures can be drawn for any molecule in which there is more than one possible position for a double bond; the number of structures will equal the number of different possible positions.

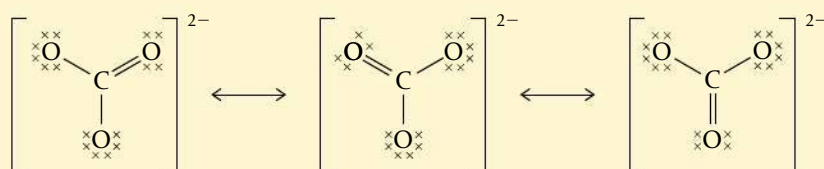
Worked example

Draw the resonance structures for the carbonate ion CO_3^{2-} .

Solution

Count the number of valence electrons: $4 + (6 \times 3) + 2 = 24$

Draw the Lewis structure, noting that there are three possible positions for the double bond. This means there will be three resonance structures, as follows:



Some other ions and molecules that have delocalized electrons and so can be represented by resonance structures are shown in the table below.

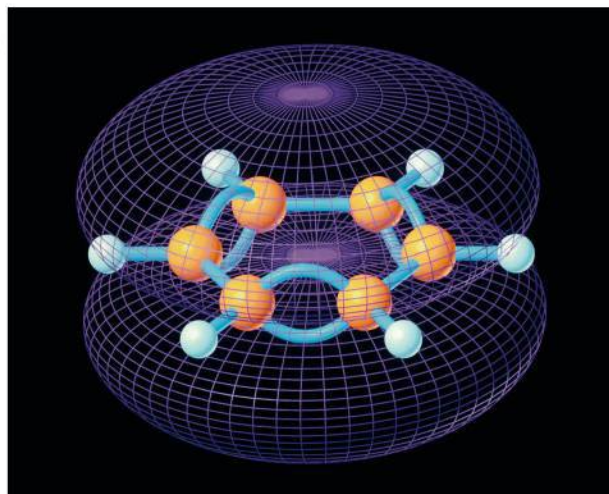
Resonance occurs when more than one valid Lewis structure can be drawn for a particular molecule. The true structure is an average of these, known as a resonance hybrid.

You can deduce how many resonance structures exist for a molecule or ion by looking at the number of possible positions for its double bond.

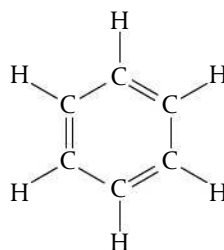
Species name and formula	Number of valence electrons	Resonance structures
nitrate(V), NO_3^-	24	$\left[\begin{array}{c} \times \times \quad \times \times \\ \text{O}=\text{N}-\text{O} \times \\ \times \times \quad \times \times \\ \\ \times \times \text{O} \times \end{array} \right]^- \longleftrightarrow \left[\begin{array}{c} \times \times \quad \times \times \\ \times \text{O}=\text{N}-\text{O} \times \\ \times \times \quad \times \times \\ \\ \times \times \text{O} \times \end{array} \right]^- \longleftrightarrow \left[\begin{array}{c} \times \times \quad \times \times \\ \times \text{O}-\text{N}=\text{O} \times \\ \times \times \quad \times \times \\ \\ \times \times \text{O} \times \end{array} \right]^-$
nitrate(III), NO_2^-	18	$\left[\begin{array}{c} \times \times \quad \times \times \quad \times \times \\ \times \text{O}-\text{N}=\text{O} \\ \times \times \quad \times \times \end{array} \right]^- \longleftrightarrow \left[\begin{array}{c} \times \times \quad \times \times \quad \times \times \\ \text{O}=\text{N}-\text{O} \times \\ \times \times \quad \times \times \end{array} \right]^-$
methanoate, HCOO^-	18	$\left[\begin{array}{c} \times \times \quad \times \times \\ \text{H}-\text{C}=\text{O} \times \\ \times \times \quad \times \times \\ \\ \times \times \text{O} \times \end{array} \right]^- \longleftrightarrow \left[\begin{array}{c} \times \times \quad \times \times \\ \text{H}-\text{C}-\text{O} \times \\ \times \times \quad \times \times \\ \\ \times \times \text{O} \times \end{array} \right]^-$

Computer artwork of the structure of benzene showing carbon atoms in orange, hydrogen atoms in white, and the delocalized bonds as purple grids above and below the plane of the molecule.

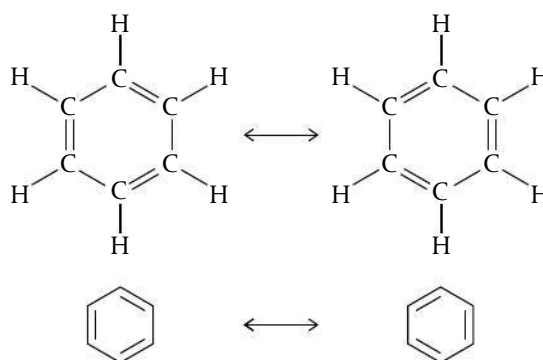
The alternating single and double bonds shown in the model could be misleading – you should understand why when you have read this section.



Benzene, C_6H_6 , is a particularly interesting case of a molecule with delocalized electrons. The six carbon atoms are arranged in a hexagonal ring, each also bonded to a hydrogen atom in a triangular planar arrangement with 120° bond angles. A possible Lewis structure is:



But we can see there are other possible positions for the double bonds, indicating the existence of resonance structures.



In Chapter 10 we will discuss evidence that indicates that benzene's properties cannot be explained by either of these Lewis structures. The true form is the resonance hybrid:



The circle inside the hexagon represents delocalized pi electrons, which are spread equally through the ring, rather than being in confined double bonds. As we would expect, this results in carbon–carbon bonds that are equal in length and strength, and intermediate between those of single and double bonds.

NATURE OF SCIENCE

Models and representations such as Lewis structures and molecular models are useful tools in science. They can help understanding and the communication of complex information effectively. But there are also often limits to the usefulness of a model. The difficulty of representing a resonance hybrid through a Lewis structure is an example of this. It demands that we extend our interpretation of the Lewis structure to include the existence of molecules with delocalized electrons.





NATURE OF SCIENCE

The current model of benzene with a symmetrical ring of delocalized electrons developed when predictions based on the so-called Kekulé model with alternating double bonds failed to match observations. Differences between the calculated and measured values of bond energy and bond length, and the ability to undergo addition reactions all indicated that the Kekulé model was not correct. Progress in science sometimes depends on the need to abandon a model or an interpretation if it cannot explain valid experimental results.



Bond order is a measure of the number of electrons involved in bonds between two atoms. Values for bond order are: single bonds = 1, double bonds = 2, triple bonds = 3. Resonance hybrids have fractional values of bond order. The carbon-carbon bonds in benzene have a bond order of 1.5.

	C–C single bond	C=C double bond	Carbon–carbon bonds in benzene
Bond length / pm	154	134	140
Bond enthalpy / kJ mol ⁻¹	346	614	507

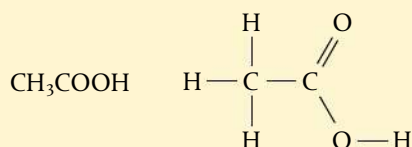
Resonance gives great stability to the benzene molecule, making it chemically quite unreactive. In our study of organic chemistry we will learn that this is unusual given its high ratio of carbon to hydrogen (unsaturation). The explanation in Chapter 10 will use concepts of hybridization and pi bonding, introduced later in this chapter.

Worked example

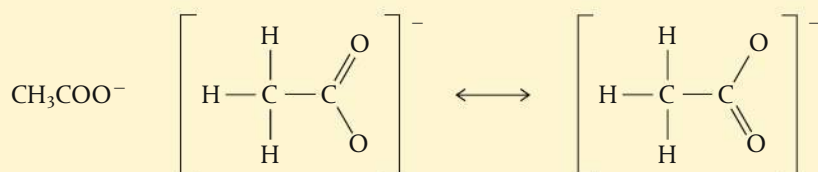
Compare the structures of CH_3COOH and CH_3COO^- with reference to their possible resonance structures.

Solution

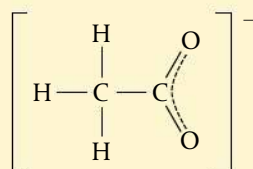
Draw the Lewis structure of each, and note if there is more than one possible position for a double bond to be placed.



There is only one possible Lewis structure, so the molecule has one carbon–oxygen double bond and one carbon–oxygen single bond.



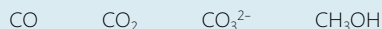
Two resonance structures exist, so the ion exists as a hybrid with two equal carbon–oxygen bonds, intermediate in length and strength between the two carbon–oxygen bonds in CH_3COOH .



Resonance is an important concept that gives special properties to the structures where it occurs. We have seen its influence on bond lengths and strengths, which in turn can influence reactivity. The different strengths of organic acids and bases, for example, can often be explained by resonance.

Exercises

23 Put the following species in order of increasing carbon–oxygen bond length:



24 By reference to their resonance structures, compare the nitrogen–oxygen bond lengths in nitrate(V) (NO_3^-) and nitric(V) acid (HNO_3).

Some covalent substances form giant molecular crystalline solids

Most covalent substances exist as discrete molecules with a finite number of atoms. However, there are some that have a very different structure, a crystalline lattice in which the atoms are linked together by covalent bonds. Effectively, the crystal is a single molecule with a regular repeating pattern of covalent bonds, so has no finite size. It is referred to as a **giant molecular** or **network covalent** structure or **macromolecular structure**. As we might expect, such crystalline structures have very different properties from other smaller covalent molecules. A few examples are considered here.

Allotropes of carbon

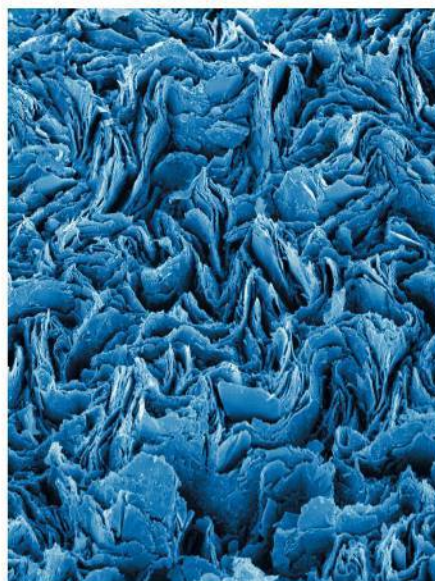
Allotropes are different forms of an element in the same physical state, such as oxygen (O_2) and ozone (O_3) which both exist as gases. Different bonding within these structures gives rise to distinct forms with different properties.

Carbon has several allotropes and these are described and compared in the table on page 169.

References to hybridization are explained in Section 14.2 of this chapter.

CHALLENGE YOURSELF

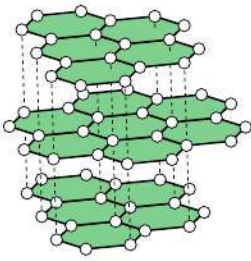
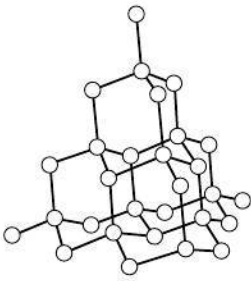
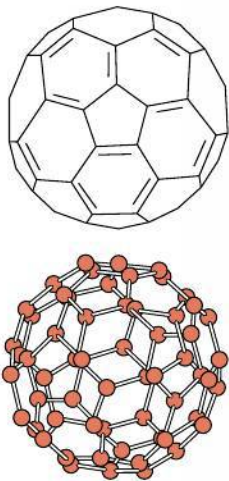
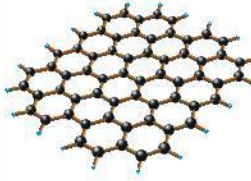
- 5** For the most part, good electrical conductors are also good thermal conductors, but this is not the case with diamond. Can you think why this might be, and whether there are other substances that are not electrical conductors but good thermal conductors?



Coloured scanning electron micrograph of layers making up the core of a graphite pencil. Because graphite is a soft form of carbon, the tip of the pencil disintegrates under pressure to leave marks on the paper.



Cut and polished diamond. Diamond is a naturally occurring form of carbon that has crystallized under great pressure. It is the hardest known mineral. Beautiful crystals are found in South Africa, Russia, Brazil, and Sierra Leone.

	Graphite	Diamond	Fullerene C ₆₀	Graphene
Structure	 Each C atom is sp^2 hybridized and covalently bonded to 3 others, forming hexagons in parallel layers with bond angles of 120°. The layers are held only by weak London dispersion forces so they can slide over each other.	 Each C atom is sp^3 hybridized and covalently bonded to 4 others tetrahedrally arranged in a regular repetitive pattern with bond angles of 109.5°.	 Each C atom is sp^2 hybridized and bonded in a sphere of 60 carbon atoms, consisting of 12 pentagons and 20 hexagons. Structure is a closed spherical cage in which each carbon is bonded to 3 others. (Note it is not a giant molecule as it has a fixed formula.)	 Each C atom is covalently bonded to 3 others, as in graphite, forming hexagons with bond angles of 120°. But it is a single layer, so exists as a two-dimensional material only. It is often described as a honeycomb or chicken wire structure.
Electrical conductivity	Good electrical conductor; contains one non-bonded, delocalized electron per atom that gives electron mobility.	Non-conductor of electricity; all electrons are bonded and so non-mobile.	A semiconductor at normal temperature and pressure due to some electron mobility; easily accepts electrons to form negative ions.	Very good electrical conductor; one delocalized electron per atom gives electron mobility across the layers.
Thermal conductivity	Not a good conductor, unless the heat can be forced to conduct in a direction parallel to the crystal layers.	Very efficient thermal conductor, better than metals.	Very low thermal conductivity.	Best thermal conductivity known, even better than diamond.
Appearance	Non-lustrous, grey crystalline solid.	Highly transparent, lustrous crystal.	Yellow crystalline solid, soluble in benzene.	Almost completely transparent.
Special properties	Soft and slippery due to slippage of layers over each other; brittle; very high melting point; the most stable allotrope of carbon.	The hardest known natural substance; it cannot be scratched by anything; brittle; very high melting point.	Very light and strong; reacts with K to make superconducting crystalline material; low melting point.	The thickness of just one atom so the thinnest material ever to exist; but also the strongest – 100 times stronger than steel; very flexible; very high melting point.
Uses	A dry lubricant; in pencils; electrode rods in electrolysis.	Polished for jewellery and ornamentation; tools and machinery for grinding and cutting glass.	Lubricants, medical, and industrial devices for binding specific target molecules; related forms are used to make nanotubes and nanobuds used as capacitors in electronics industry, and catalysts.	TEM (transmission electron microscopy) grids, photo-voltaic cells, touch screens, high performance electronic devices; many applications are still being developed.

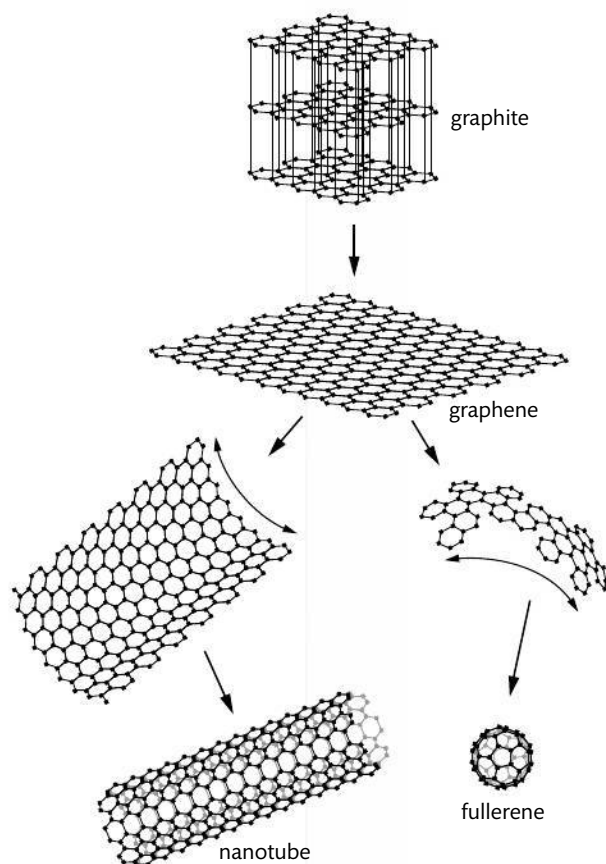
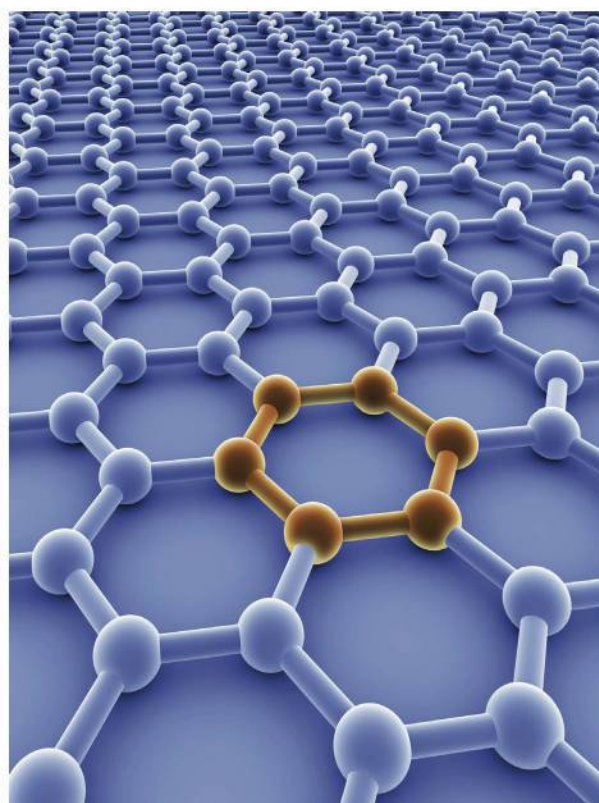


Figure 4.10 Graphite occurs naturally, and a single separated layer is graphene. A rolled up layer of graphene is a nanotube, and a closed cage is fullerene.



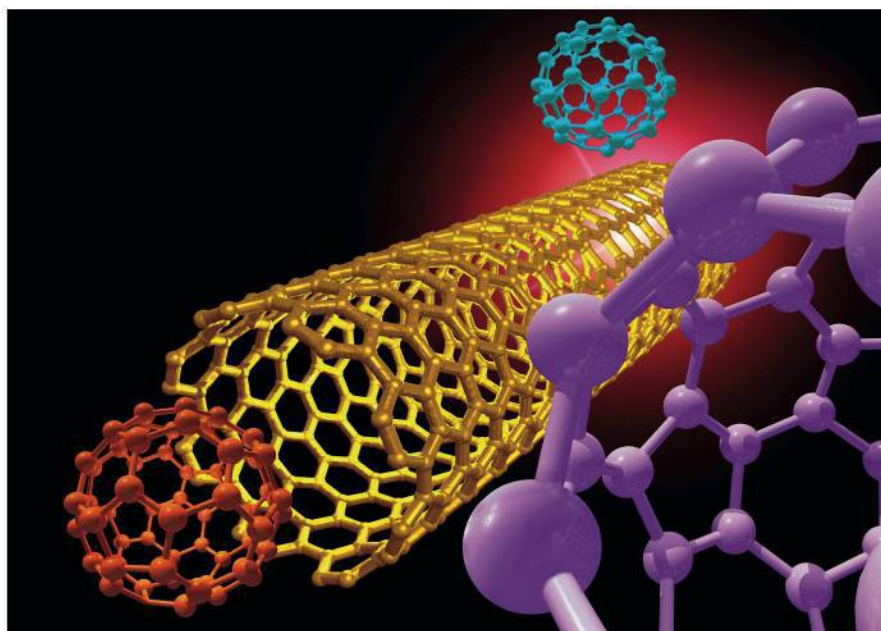
Graphene is a planar sheet of carbon atoms arranged in a hexagonal pattern.

Graphene is a relatively new form of carbon, isolated for the first time in 2004. Given its hardness, strength, lightness, flexibility, conductivity, and chemical inertness, it has enormous potential applications. Currently though, many of these are only in the development stage as they are dependent on the production of large sheets of pure graphene, which is difficult to achieve economically. But this is an intense field of research and breakthroughs are common. Some of the potential applications and advantages of graphene are as follows:

- new conductive materials could be made by mixing graphene with plastics;
- transistors of graphene are predicted to be faster and smaller than silicon transistors in electronics;
- touch screens of graphene printed on thin plastic instead of coatings layered onto glass would be light, flexible, and almost unbreakable;
- supercapacitors for storing charge could replace batteries and more efficiently power mobile devices and electric cars;
- solar panels in which photovoltaic cells (that generate electricity from sunlight) contain graphene would be light and flexible, and so could be wrapped around any surface such as clothing, furniture, or vehicles.



Whether graphene will realize its potential and lead to innovative materials and applications remains an open question. What is the role of imagination in helping to direct the research focus of scientists?



Computer artwork of spherical and cylindrical fullerenes – buckyballs and carbon nanotubes. These substances are being investigated for a wide range of technical and medical uses. Here the carbon nanotubes have been engineered to have a diameter just large enough to allow buckyballs to pass through them.

NATURE OF SCIENCE

The discoveries of the allotropes of carbon – buckminsterfullerene and graphene – illustrate the importance of collaboration between scientists.

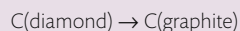
The discovery of fullerenes in 1985 was the result of teamwork between scientists with different experience and research objectives. Harold Kroto from the UK was interested in red giant stars and how carbon might polymerize near them. Robert Curl and Richard Smalley working in Texas, USA, had developed a technique using a laser beam for evaporating and analysing different substances. When they worked together and applied this technique to graphite, clusters of stable C_{60} and C_{70} spheres were formed. The three scientists shared the Nobel Prize in Chemistry for 1996.

The isolation of graphene was announced by Kostya Novoselov and Andre Geim in 2004, both working at the University of Manchester, UK. It led to their shared award of the Nobel Prize in Physics in 2010. Novoselov and Geim were research collaborators over many years in different countries, originally as research student and supervisor respectively. Their work involved the use of adhesive tape to rip successively thinner flakes off graphite and analyse them by attachment to a layer of silicon oxide. Many scientists over many years had attempted to isolate graphene, and through their successes and failures indirectly contributed to the later discovery.

The development of structures such as nanotubes, nanobuds, and graphene is part of the growing science of **nanotechnology** (Chapter 12), which deals with the atomic scale manipulation of matter. An interesting development is a new material called **graphone**, which is made by adding hydrogen to graphene. Controlling the amount of hydrogen coverage gives variable magnetic properties, which may in turn lead to wider graphene applications. A further development is fluorographene, which has one fluorine atom attached to each carbon atom giving a rippled rather than a flat structure. As all the electrons are held in bonds, it is an insulator.

CHALLENGE YOURSELF

- 6 Graphite is thermodynamically more stable than diamond. The reaction



is accompanied by a loss of energy, $\Delta H = -2 \text{ kJ mol}^{-1}$.

But we commonly hear that 'diamonds are forever'. Are they?



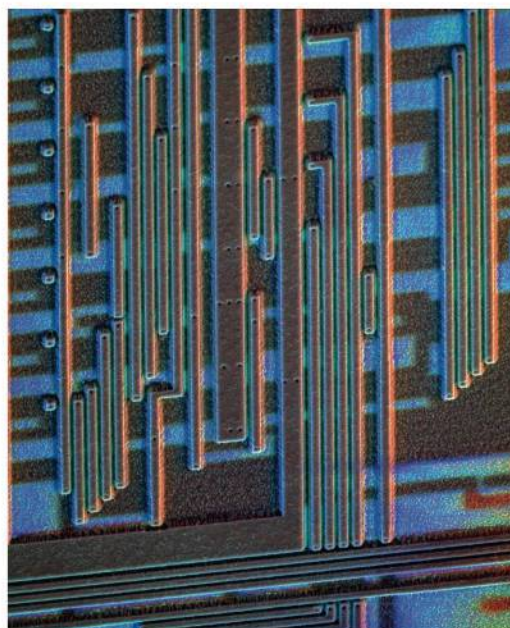
The name 'fullerene' was given to the newly discovered spheres C_{60} in honour of the American architect R. Buckminster Fuller. He had designed the World Exhibition building in Montreal, Canada, on the same concept of hexagons and a small number of pentagons to create a curved surface known as a geodesic dome. The dome of the Epcot Centre in Disney World, Florida, is similarly designed. Perhaps more familiarly it is also the structure of a European soccer ball. The term 'buckyballs' has slipped into common usage, derived from the full name of the structure 'buckminsterfullerene'.



The structure of fullerene has the same arrangements of hexagons and pentagons as a soccer ball.

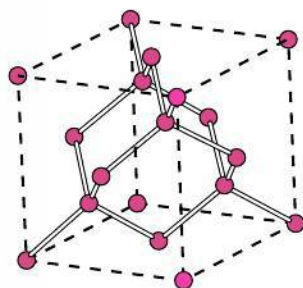
Silicon is the most abundant element in the Earth's crust after oxygen, occurring as silica (SiO_2) in sand and in the silicate minerals. Since silicon is just below carbon in Group 14, the possibility of silicon-based life has been proposed. But unlike carbon, silicon is not able to form long chains, multiple bonds, or rings so cannot compete with the diversity possible in organic chemistry, based on carbon. However, there is some evidence that the first forms of life were forms of clay minerals that were probably based on the silicon atom.

Silicon and silicon dioxide

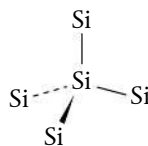


Coloured scanning electron micrograph of the surface of a microprocessor computer chip. Silicon chips are tiny pieces of silicon as small as 1 mm^2 made to carry minute electrical circuits used in computers and transistors. Silicon is the most widely used semiconductor.

Like carbon, silicon is a Group 14 element and so its atoms have four valence shell electrons. In the elemental state, each silicon atom is covalently bonded to four others in a tetrahedral arrangement. This results in a giant lattice structure much like diamond.



The silicon crystal structure and the arrangement of bonds around a central silicon atom.



Silicon dioxide, SiO_2 , commonly known as silica or quartz, also forms a giant covalent structure based on a tetrahedral arrangement. But here the bonds are between silicon and oxygen atoms, where each Si atom is covalently bonded to four O atoms, and each O to

two Si atoms. You can think of the oxygen atoms as forming bridges between the tetrahedrally bonded silicon atoms.



Quartz crystals shown in a coloured scanning electron micrograph. Quartz is a form of silica (SiO_2) and the most abundant mineral in the Earth's crust. Quartz is used in optical and scientific instruments and in electronics such as quartz watches.

Note here the formula SiO_2 refers to the *ratio* of atoms within the giant molecule – it is an empirical formula and the actual number of atoms present will be a very large multiple of this. As the atoms are strongly held in tetrahedral positions that involve all four silicon valence electrons, the structure has the following properties:

- strong;
- insoluble in water;
- high melting point;
- non-conductor of electricity.

These are all properties we associate with glass and sand – different forms of silica.

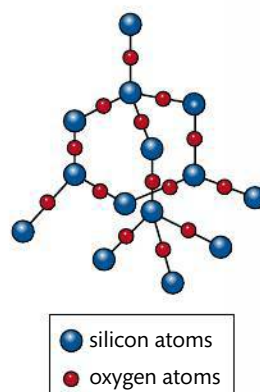


Figure 4.11 The structure of quartz SiO_2 .

Exercises

- 25** Describe the similarities and differences you would expect in the properties of silicon and diamond.
- 26** Explain why graphite and graphene are good conductors of electricity whereas diamond is not.

4.4 Intermolecular forces

Understandings:

- Intermolecular forces include London (dispersion) forces, dipole–dipole forces, and hydrogen bonding.

Guidance

The term ‘London (dispersion) forces’ refers to instantaneous dipole–induced dipole forces that exist between any atoms or groups of atoms and should be used for non-polar entities. The term ‘van der Waals’ is an inclusive term, which includes dipole–dipole, dipole–induced dipole, and London (dispersion) forces.

- The relative strengths of these interactions are London (dispersion) forces < dipole–dipole forces < hydrogen bonds.

Applications and skills:

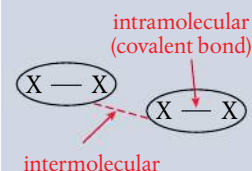
- Deduction of the types of intermolecular force present in substances, based on their structure and chemical formula.
- Explanation of the physical properties of covalent compounds (volatility, electrical conductivity, and solubility) in terms of their structure and intermolecular forces.

Covalent bonds hold atoms together *within* molecules, but of course molecules do not exist in isolation. A gas jar full of chlorine, Cl_2 , for example, will contain millions of molecules of chlorine. So what are the forces that exist *between* these molecules, the so-called **intermolecular forces**? The answer depends on the polarity and size of the molecules involved, and so the intermolecular forces will vary for different molecules. We will consider three types of intermolecular force here, and see how they differ from each other in origin and in strength.

The strength of intermolecular forces determines the physical properties of a substance. Volatility, solubility, and conductivity can all be predicted and explained from knowledge of the nature of the forces between molecules.



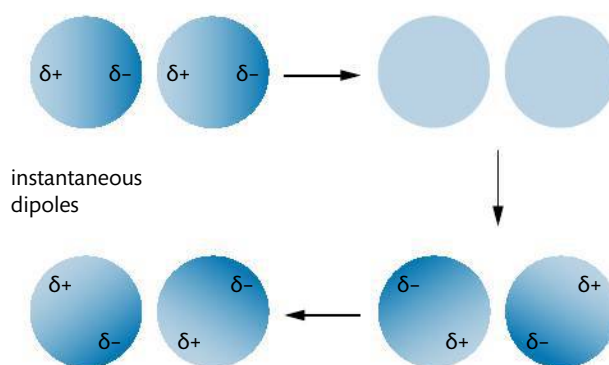
Note that the prefix *intra-* refers to *within*, whereas the prefix *inter-* refers to *between*. For example, the Internet gives communication between many different places, while an intranet works within an organization.



Fritz London (1900–1954) was a German physicist who pioneered the interpretation of covalent bond formation and intermolecular forces in the emerging field of quantum mechanics. As he was of Jewish descent, he had to leave his university post in Germany during the Nazi regime before World War II, and completed most of his research in Duke University, USA.

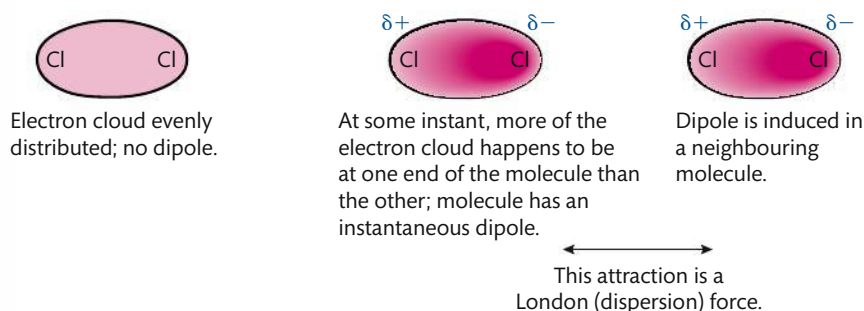


Figure 4.12 A series of instantaneous views of electron density in an atom. Each image is like a snapshot of a possible electron distribution at a moment in time. Although the instantaneous dipole varies, it always leads to an attractive force with its neighbour.



As a result, weak forces of attraction, known as **London (dispersion) forces**, will occur between opposite ends of these two temporary dipoles in the molecules. These are the weakest form of intermolecular force. Their strength increases with increasing molecular size; this is because the greater number of electrons within a molecule increases the probability of temporary dipoles developing.

Figure 4.13 Instantaneous dipole-induced dipole cause London (dispersion) forces between molecules of Cl_2 .



London (dispersion) forces are the *only* forces that exist between non-polar molecules. Such molecules generally have low melting and boiling points, because relatively little energy is required to break the weak London (dispersion) forces and separate the molecules from each other. This is why many non-polar elements and compounds are gases at room temperature, for example O_2 , Cl_2 , and CH_4 . Boiling point data also indicate how the strength of London (dispersion) forces increases with increasing molecular size. This is shown in the tables on page 175 that compare the boiling points of the halogens (Group 17 elements) and of the family of hydrocarbons known as the alkanes.

Element	M_r	Boiling point / °C	State at room temperature
F_2	38	-188	gas
Cl_2	71	-34	gas
Br_2	160	59	liquid
I_2	254	185	solid

boiling point increases with increasing number of electrons

Alkane	M_r	Boiling point / °C
CH_4	16	-164
C_2H_6	30	-89
C_3H_8	44	-42
C_4H_{10}	58	-0.5

boiling point increases with increasing number of electrons

Although London (dispersion) forces are weak, they are responsible for the fact that non-polar substances can be condensed to form liquids and sometimes solids at low temperatures. For example, nitrogen gas, N_2 , liquefies below $-196^\circ C$, and even helium gas has been liquefied at the extremely low temperature of $-269^\circ C$.

London (dispersion) forces are also components of the forces between polar molecules, but often get somewhat overlooked because of the presence of stronger forces.

Dipole–dipole attraction

Polar molecules such as hydrogen chloride, HCl , have a permanent separation of charge within their bonds as a result of the difference in electronegativity between the bonded atoms. One end of the molecule is electron deficient with a partial positive charge ($\delta+$), while the other end is electron rich with a partial negative charge ($\delta-$). This is known as a **permanent dipole**. It results in opposite charges on neighbouring molecules attracting each other, generating a force known as a **dipole–dipole attraction**.

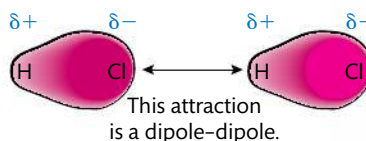


Figure 4.14 Permanent dipoles cause forces of dipole–dipole attraction between molecules of HCl .

The strength of this intermolecular force will vary depending on the distance and relative orientation of the dipoles. But dipole–dipole forces are generally stronger than London dispersion forces, as we might expect from their origin in permanent rather than instantaneous dipoles. These forces cause the melting and boiling points of polar compounds to be higher than those of non-polar substances of comparable molecular mass.

	C_3H_8	CH_3OCH_3
Molar mass / $g\ mol^{-1}$	44	46
Intermolecular attraction	London dispersion forces	London dispersion forces and dipole–dipole attraction
Boiling point / K	229	249

When examining the boiling points of different substances to see the effect of different intermolecular forces, it is important to compare substances with similar molecular mass. Otherwise the difference could be attributed solely to stronger London dispersion forces in the molecules with more electrons.

For two substances of similar molecular mass, the more polar substance will have the higher boiling point.

Johannes van der Waals (1837–1923) from the Netherlands established himself as an eminent physicist on the publication of his very first paper, his PhD thesis. His study of the continuity of the gas and liquid state from which he put forward his 'equation of state' led James Clerk Maxwell to comment, 'there can be no doubt that the name of van der Waals will soon be among the foremost in molecular science'. van der Waals did indeed fulfil this early promise, being awarded the Nobel Prize in Physics in 1910.

Figure 4.15 Hydrogen bonding between water molecules.

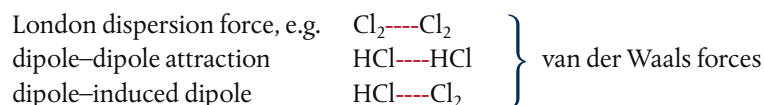
There is ongoing debate about the nature of the hydrogen bond and IUPAC define six criteria that should be used as evidence for its occurrence. To what extent does a specialized vocabulary help or hinder the growth of knowledge?

Hydrogen bonding only occurs between molecules in which hydrogen is bonded directly to fluorine, nitrogen, or oxygen.

Figure 4.16 Periodic trends in the boiling points of the hydrides of Groups 14–17.

Note that dipole–dipole attractions can occur between any combinations of polar molecules, such as between PCl_3 and CHCl_3 , and generally lead to the solubility of polar solutes in polar solvents.

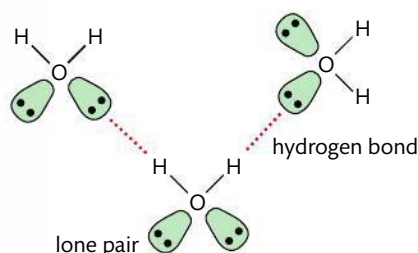
The umbrella term **van der Waals' forces** is used to include both London dispersion forces and dipole–dipole attractions. It also covers a less common type of attraction known as dipole–induced dipole. In other words, van der Waals' forces refers to all forces between molecules that do not involve electrostatic attractions between ions or bond formation. We can summarize this as follows:



In some cases, van der Waals' forces may occur within a molecule (intramolecularly) if the different groups can position themselves appropriately. This is an important feature of proteins, for example, which is discussed in detail in Chapter 13.

Hydrogen bonding

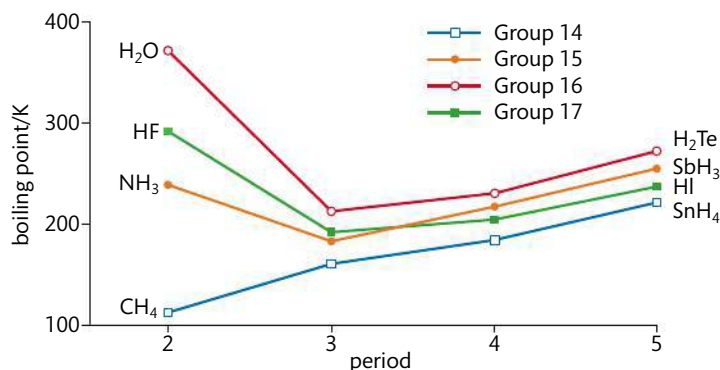
When a molecule contains hydrogen covalently bonded to a very electronegative atom (fluorine, nitrogen, or oxygen), these molecules are attracted to each other by a particularly strong type of intermolecular force called a **hydrogen bond**. The



hydrogen bond is in essence a particular case of dipole–dipole attraction. The large electronegativity difference between hydrogen and the bonded fluorine, oxygen, or nitrogen results in the electron pair being pulled away from the hydrogen. Given its small size and the fact that it has no other electrons to shield the nucleus, the hydrogen now exerts a strong

attractive force on a lone pair in the electronegative atom of a neighbouring molecule. This is the hydrogen bond.

Hydrogen bonds are the strongest form of intermolecular attraction. Consequently, they cause the boiling points of substances that contain them to be significantly higher than would be predicted from their molar mass. We can see this in Figure 4.16, in which boiling points of the hydrides from Groups 14 to 17 are compared down the Periodic Table.



In all four groups there is a general trend that boiling point increases down the group as molar mass increases. The anomalies are NH_3 , HF , and H_2O , which all have significantly higher boiling points than would be expected from their molar mass. This can only be explained by the presence of hydrogen bonding in these molecules. If it wasn't for the fact that it is hydrogen bonded, H_2O would be a gas not a liquid at room temperature.

Likewise, when we compare the boiling points of some organic molecules that have similar or equal values of molar mass, we find a higher value where hydrogen bonding occurs between the molecules. For example, the following table compares two different forms of $\text{C}_2\text{H}_6\text{O}$, known as isomers, both with a molar mass of 46.

$\text{CH}_3\text{-O-CH}_3$	$\text{CH}_3\text{CH}_2\text{-O-H}$
methoxymethane	ethanol
$M = 46 \text{ g mol}^{-1}$	$M = 46 \text{ g mol}^{-1}$
does not form hydrogen bonds	forms hydrogen bonds
boiling point = -23°C	boiling point = $+79^\circ\text{C}$

Water makes a particularly interesting case for the study of hydrogen bonding. Here, because of the two hydrogen atoms in each molecule and the two lone pairs on the oxygen atom, each H_2O can form up to *four* hydrogen bonds with neighbouring molecules. Liquid water contains fewer than this number, but in the solid form, ice, each H_2O is maximally hydrogen bonded in this way. The result is a tetrahedral arrangement that holds the molecules a fixed distance apart, forming a fairly open structure, which is actually *less* dense than the liquid. This is a remarkable fact – in nearly all other substances the solid form with closer packed particles is *more* dense than its liquid. The fact that ice floats on water is evidence of the power of hydrogen bonds in holding the molecules together in ice. This density change means that water expands on freezing, which can lead to all kinds of problems in cold climates such as burst pipes and cracked machinery. The same force of expansion is at work in the Earth's crust, fragmenting and splitting rocks, ultimately forming sand and soil particles. The humble hydrogen bond is truly responsible for massive geological changes!

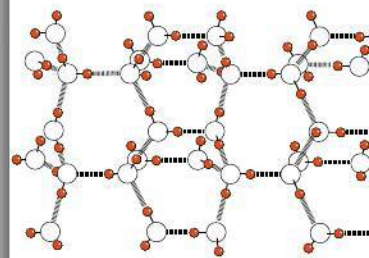


Figure 4.17 The arrangement of water molecules in ice. Each molecule is held by hydrogen bonds to four other molecules in a tetrahedral arrangement.

Icebergs float on water. Hydrogen bonds create a very open structure in ice which is less dense than liquid water.



Make sure you realize that although hydrogen bonds are strong in relation to other types of intermolecular force, they are very much weaker (about 15–20 times) than covalent bonds and ionic bonds.



Computer artwork of a DNA molecule replicating. DNA is composed of two strands, held together by hydrogen bonds and twisted into a double helix. Before replication the strands separate from each other by breaking the hydrogen bonds and each strand then acts as a template for the synthesis of a new molecule of DNA.

NATURE OF SCIENCE

The relationship between the intermolecular forces that exist in a substance and its physical properties is a good example of the predictive power of science. This is applied in the development of new materials, such as in the petrochemicals and pharmaceuticals industries. For example, being able to predict properties such as solubility and melting point from theoretical structures is crucial in drug design.

Make sure you realize that when a covalent substance melts or boils, it is the bonds *between* the molecules that break, not the covalent bonds *within* the molecules. When you boil a kettle of water, you do not fill the kitchen with separated oxygen and hydrogen atoms.



		Non-polar molecules	Polar molecules	Molecules which contain H-O, H-N, or H-F
London dispersion forces	} van der Waals' forces	✓	✓	✓
Dipole-dipole attractions			✓	✓
Hydrogen bonding				✓

increasing strength of intermolecular force

In these examples we have looked at hydrogen bonds as an intermolecular force, that is, *between* molecules, but like van der Waals' forces, they can occur *within* large molecules where they play a key role in determining properties. Proteins, for example, are fundamentally influenced by hydrogen bonding – you can read about this in Chapter 13, Option B Biochemistry. Another fascinating example is DNA (deoxyribonucleic acid) which, as the chemical responsible for storing the genetic information in cells, is able to replicate itself exactly, a feat only possible because of its use of hydrogen bonding.

We can now summarize the intermolecular forces we have discussed in this section.

The physical properties of covalent compounds are largely a result of their intermolecular forces

If we know the type of intermolecular force in a substance, we are in a good position to predict its physical properties.

Melting and boiling points

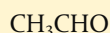
Changing state by melting or boiling both involve separating particles by overcoming the forces between them. So the stronger the intermolecular forces, the more energy will be required to do this, and the higher will be the substance's melting and boiling points.

Covalent substances generally have lower melting and boiling points than ionic compounds. The forces to be overcome to separate the molecules are the relatively weak intermolecular forces, which are significantly easier to break than the electrostatic attractions in the ionic lattice. This is why many covalent substances are liquids or gases at room temperature.

As discussed on page 174, the strength of the intermolecular forces increases with increasing molecular size and with an increase in the extent of polarity within the molecule, and so these features help us to predict relative melting and boiling points.

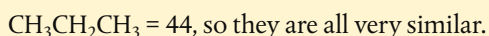
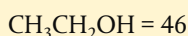
Worked example

Put the following molecules in order of increasing boiling point and explain your choice:

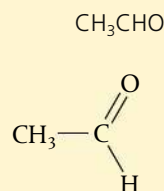


Solution

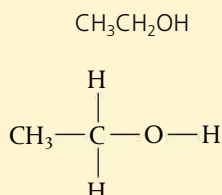
First check the value of M for each molecule:



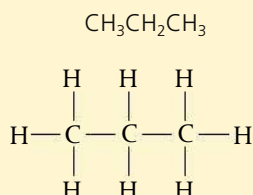
Now consider the type of bonding and intermolecular attractions:



polar bonds $\Rightarrow$
dipole-dipole

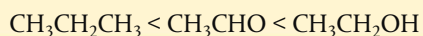


O-H bond $\Rightarrow$ H bonding



non-polar bonds $\Rightarrow$
London (dispersion) forces

So the order, starting with the lowest boiling point, will be:



By contrast, macromolecular or giant covalent structures have high melting and boiling points, often at least as high as those of ionic compounds. This is because covalent bonds must be broken in these compounds for the change of state to occur. Diamond, for example, remains solid until about 4000 °C.

Solubility

Non-polar substances are generally able to dissolve in non-polar solvents by the formation of London dispersion forces between solute and solvent. For example, the halogens, all of which are non-polar molecules (e.g. Br_2), are readily soluble in the non-polar solvent paraffin oil. It is another example of *like dissolves like*, which is a useful guiding principle here.

Likewise, polar covalent compounds are generally soluble in water, a highly polar solvent. Here the solute and solvent interact through dipole interactions and hydrogen bonding. Common examples include the aqueous solubility of HCl, glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), and ethanol ($\text{C}_2\text{H}_5\text{OH}$). Biological systems are mostly based on polar covalent molecules in aqueous solution.

The solubility of polar compounds is, however, reduced in larger molecules where the polar bond is only a small part of the total structure. The non-polar parts of the molecule, unable to associate with water, reduce its solubility. For example, while ethanol ($\text{C}_2\text{H}_5\text{OH}$) is readily soluble in water, the larger alcohol heptanol ($\text{C}_7\text{H}_{15}\text{OH}$) is

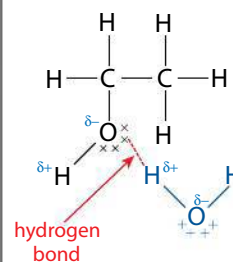


Figure 4.18 Water and ethanol form hydrogen bonds and mix readily, as we know from the homogeneous nature of alcoholic drinks.

The moss plant *Philonotis fontana* has a waxy substance on its surface which repels water. The wax is a large non-polar covalent molecule that is unable to form hydrogen bonds with water.

Dry-cleaning is a process where clothes are washed without water. Instead, a less polar organic liquid is used which may be a better solvent for stains caused by large non-polar molecules such as grass and grease.

Non-polar covalent substances dissolve best in non-polar solvents, polar substances dissolve best in polar solvents.



not. The difference is the size of the non-polar hydrocarbon group.

It follows from this that non-polar covalent substances do not dissolve well in water. Nitrogen, N_2 , for example, has very low solubility in water at normal pressure, as do hydrocarbons such as candle wax and large covalent molecules such as chlorophyll. If you think about the substances that mark clothes and are not easily removed with water – grease, red wine, grass stains, and gravy are good examples – they are all relatively large insoluble non-polar molecules.

Likewise, polar substances have low solubility in non-polar solvents as they will remain held to each other by their dipole–dipole attractions and cannot interact well with the solvent. You can observe this by putting some sugar into cooking oil, and noticing that it does not dissolve.

Giant molecular substances are generally insoluble in all solvents, as too much energy would be required to break the strong covalent bonds in their structure. Beaches made primarily of sand, SiO_2 , do not dissolve when the tide comes in and a diamond ring can safely be immersed in water.

Electrical conductivity

Covalent compounds do not contain ions, and so are not able to conduct electricity in the solid or liquid state. Some polar covalent molecules, however, in conditions where they can ionize will conduct electricity. For example, HCl dissolved in water (hydrochloric acid) is an electrical conductor as it dissociates into $H^+(aq) + Cl^-(aq)$.

As discussed on page 169, the giant covalent molecules graphite and graphene are electrical conductors due to mobile electrons. In addition, fullerene and silicon have semiconductivity properties. Diamond is a non-conductor of electricity.

Summary of physical properties

We can now summarize the physical properties of ionic and covalent compounds, though it should be recognized that the table below gives only general trends and does not apply universally.

	Ionic compounds	Polar covalent compounds	Non-polar covalent compounds	Giant covalent
Volatility	low	higher	highest	low
Solubility in polar solvent, e.g. water	soluble	solubility increases as polarity increases	non-soluble	non-soluble
Solubility in non-polar solvent, e.g. hexane	non-soluble	solubility increases as polarity decreases	soluble	non-soluble
Electrical conductivity	conduct when molten (l) or dissolved in water (aq)	non-conductors	non-conductors	non-conductors except graphite, graphene and semi-conductivity of Si and fullerene

Exercises

27 The physical properties of five solids labelled A, B, C, D, and E are summarized below. The substances are: an ionic compound, a non-polar molecular solid, a metal, a polar molecular solid, and a giant molecular substance. Classify each correctly.

Sample	Solubility in water	Conductivity of solution	Conductivity of solid	Relative melting point
A	insoluble	–	yes	third to melt
B	insoluble	–	no	highest
C	soluble	no	no	second to melt
D	insoluble	–	no	lowest
E	soluble	yes	no	fourth to melt

28 Which substance is the most soluble in water?

- A** CH_3OH **B** CH_4 **C** C_2H_6 **D** $\text{C}_2\text{H}_5\text{OH}$

29 State the intermolecular forces that exist between molecules of each of the following:

- (a)** dry ice, $\text{CO}_2(\text{s})$ **(b)** $\text{NH}_3(\text{l})$ **(c)** $\text{N}_2(\text{l})$ **(d)** CH_3OCH_3

30 Which of each pair has the lower boiling point?

- (a)** C_2H_6 and C_3H_8 **(b)** H_2O and H_2S **(c)** Cl_2 and Br_2 **(d)** HF and HCl



Investigation of the physical properties of a variety of substances

Full details of how to carry out this experiment with a worksheet are available online.

4.5 Metallic bonding

Understandings:

- A metallic bond is the electrostatic attraction between a lattice of positive ions and delocalized electrons.
- The strength of a metallic bond depends on the charge of the ions and the radius of the metal ion.
- Alloys usually contain more than one metal and have enhanced properties.

Guidance

Examples of various alloys should be covered.

Applications and skills:

- Explanation of electrical conductivity and malleability in metals.
- Explanation of trends in melting points of metals.

Guidance

Trends should be limited to s- and p-block elements.

- Explanation of the properties of alloys in terms of non-directional bonding.

Metals are found on the left of the Periodic Table and have a small number of electrons in their outer shell. These are atoms with low ionization energies, and typically react with other elements by losing their valence electrons and forming positive ions. So these elements are characterized by having a loose control over their outer shell electrons.

In the elemental state, when there is no other element present to accept the electrons and form an ionic compound, the outer electrons are held only loosely by the metal atom's nucleus and so tend to 'wander off' or, more correctly, become **delocalized**. As we saw in Section 4.3, delocalized electrons are not fixed in one position. This means that in metals they are no longer associated closely with any one metal nucleus but instead can spread themselves through the metal structure. The metal atoms without

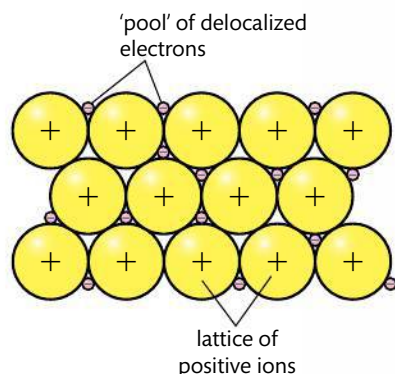


Figure 4.19 A model of metallic bonding. It is often described as a lattice of cations surrounded by a sea of electrons.

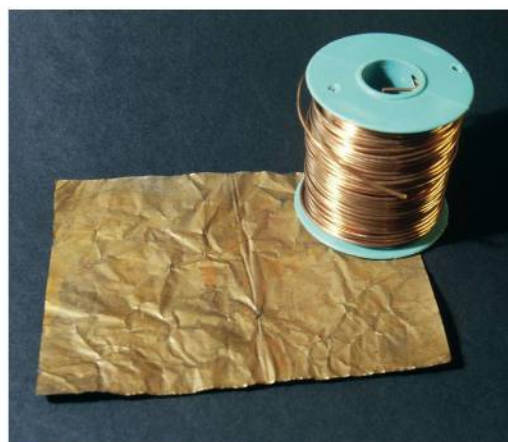
Copper is one of the oldest metals ever used. It is malleable, ductile, and a good conductor of heat and electricity. Brass and bronze are copper alloys.

Metallic bonds are the strong forces of attraction between delocalized electrons and positive ions.

The extraction of metals from their ores is a major part of industry and development in many countries, but is associated with some complex environmental issues. These include the destruction of landscapes by strip mining, the 'tailings' or waste material that accumulate in spoil tips, and the release of toxic materials that leach out of waste and pollute the environment. An important step to reduce the impact of these effects is to maximize the reuse and recycling of all metal objects, and avoid metal dumping. Despite their apparent abundance, metals are a precious resource, available only at considerable cost.

these electrons have become positively charged ions and form a regular lattice structure through which these electrons can move freely.

You can think of it like a close neighbourhood of families where the children do not belong specifically to any one set of parents but are free to wander between the homes. This arrangement causes a close association between the families. Likewise in metals there is a force of electrostatic attraction between the lattice of cations and the delocalized electrons, and this is known as **metallic bonding**.



The strength of the metallic bond is determined by:

- the number of delocalized electrons;
- the charge on the cation;
- the radius of the cation.

The greater the number of delocalized electrons and the smaller the cation, the greater the binding force between them. For example, if we compare:

- sodium, Na, Group 1, electron configuration $1s^2 2s^2 2p^6 3s^1$ and
- magnesium, Mg, Group 2, electron configuration $1s^2 2s^2 2p^6 3s^2$

we can deduce that Na will have one delocalized electron per atom and the Na^+ ion, whereas Mg will have two delocalized electrons per atom and the Mg^{2+} ion. In addition, the radius of Mg^{2+} is slightly smaller than that of Na^+ . These factors all indicate that Mg has stronger metallic bonding than Na, which can be confirmed by a comparison of their melting points.

	Na	Mg
Melting point / °C	98	650

The strength of metallic bonding tends to decrease down a group as the size of the cation increases, reducing the attraction between the delocalized electrons and the positive charges. Again melting points confirm this.

	Na	K	Rb
Melting point / °C	98	63	39

The transition elements tend to have very strong metallic bonds due to the large number of electrons that can become delocalized, from both 3d and 4s sub-shells.

Metals share characteristic physical properties, which can be explained through this model of metallic bonding. They are summarized below.

Metallic property	Explanation	Application
good electrical conductivity	delocalized electrons are highly mobile, and so can move through the metal structure in response to an applied voltage	electrical circuits use copper
good thermal conductivity	delocalized electrons and close packed ions enable efficient transfer of heat energy	cooking utensils
malleable, can be shaped under pressure	movement of delocalized electrons is non-directional and essentially random through the cation lattice, so the metallic bond remains intact while the conformation changes under applied pressure	moulded into many forms including machinery and structural components of buildings and vehicles
ductile, can be drawn out into threads		electric wires and cables
high melting points	a lot of energy is required to break the strong metallic bonds and separate the atoms	high speed tools and turbine engines; tungsten has the highest melting point
shiny, lustrous appearance	delocalized electrons in metal crystal structure reflect light	ornamental structures

NATURE OF SCIENCE

Atomic theory enables scientists to explain the distinct properties of metals. This is applied in research in the development of new materials such as alloys that combine the characteristics of different metals. The ability to explain natural phenomena such as metallic properties through the application of theory is an important feature of science.



Alloys are solutions of metals with enhanced properties

Alloys are produced by adding one metal element to another metal (or carbon) in the molten state, so that the different atoms can mix. As the mixture solidifies, ions of the different metals are scattered through the lattice and bound by the delocalized electrons, so they contain metallic bonds. The production of alloys is possible because of the non-directional nature of the delocalized electrons, and the fact that the lattice can accommodate ions of different size.

Alloys have properties that are distinct from their component elements due to the different packing of the cations in the lattice. The alloy is often more chemically stable, and also often stronger and more resistant to corrosion.

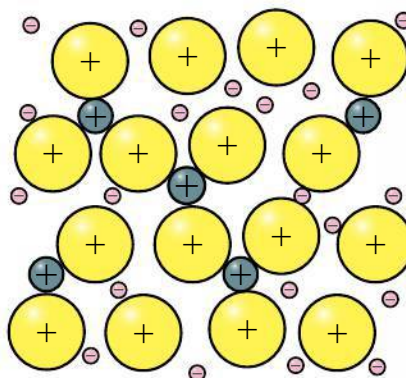


Figure 4.20 Alloys consist of different metal ions and a sea of delocalized electrons. The smaller ions are able to fit in the spaces between the larger ions in the lattice structure.



Alloys are solid solutions usually containing more than one metal, and held together by metallic bonding.